

The role of thermodynamic forcing in direct numerical simulations of turbulence-cloud microphysics interactions

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Particle-based direct numerical simulation (DNS) is a promising approach to study microscale turbulence-cloud-aerosol interactions. Such a model requires external forcing to sustain turbulent fluctuations in the momentum and thermodynamic fields. Although momentum forcing is common in particle-based DNS studies of atmospheric clouds, thermodynamic forcing has not gained enough attention. This study implements external forcing in the thermodynamic fields of temperature and vapor mixing ratio in addition to momentum forcing. The forcing terms are constructed in the Fourier space using a band of low wavenumbers. The turbulent flow of air is assumed to be homogeneous and isotropic. Aerosol particles and cloud droplets are tracked in Lagrangian fashion, with processes of condensation, evaporation, activation, and deactivation considered together. The impacts of thermodynamic forcing on entrainment of ambient air into the cloud and subsequent turbulent mixing are examined. It is found that thermodynamic forcing sustains fluctuations in the thermodynamic fields and reduces the mean temperature, water vapor mixing ratio, and supersaturation. Although thermodynamic forcing is applied at large scales only, it influences the energy at all scales by interacting with latent heat change. The probability distributions of forced thermodynamic fields are negatively skewed with kurtosis values larger than three. Further analysis of microphysical properties shows that thermodynamic forcing enhances evaporation and deactivation with broadening of particle size distributions. The activation and deactivation regimes are fluctuation-dominated if thermodynamic forcing is present as opposed to mean-dominated only without thermodynamic forcing.

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NOMENCLATURE

c_p, C_d	= specific heat of air [J/(kg.K)], time rate of exchange of mass [1/s]
δ_{k,k_f}, η	= delta function [-], Kolmogorov length scale [m]
$\epsilon_{T,in}, \epsilon_{qv,in}$	= dissipation rates in the initial temperature [K ² /s], and vapor [(g/kg) ² /s] fields
ϵ	= turbulent kinetic energy dissipation rate [m ² /s ³]
$\mathbf{F}_b, \mathbf{F}_c$	= buoyancy force [m/s ²], external force [m/s ²] in the momentum field per unit mass
F_T, F_{qv}	= external forcing in the temperature [K/s], water vapor [(g/kg)/s] fields
$\mathbf{g}, \mathbf{k}_f$	= acceleration vector due to gravity [m/s ²], forcing wavenumber vector [1/m]
k_{min}, k_{max}	= minimum wavenumber [1/m], maximum wavenumber [1/m]
κ, k_T	= hygroscopicity of solute [-], thermal conductivity in air [W/(m.K)]
l_0, l	= length scale characterizing large eddies [m], integral length scale [m]
L_h, λ, M_l	= specific latent heat [J/kg], Taylor microscale [m], molar mass of water [kg/mol]
m_a, μ_T, μ_v	= mass of air [kg], molecular diffusivities [m ² /s] of temperature, water vapor in air
ν, p	= kinematic viscosity of air [m ² /s], instantaneous fluid pressure [N/m ²]
$q_v, q_{v,s}$	= instantaneous water vapor mixing ratio [g/kg], saturation vapor mixing ratio [g/kg]
r_d, r	= dry aerosol radius [μ m], wet radius of aerosol particle or cloud droplet [μ m]
r_c, R_v	= critical radius [μ m], specific gas constant for water vapor [J/(kg.K)]
$R_{q'_v(F_{qv}-C_d)'}$	= correlation coefficient between fluctuations in water vapor and source term [-]
$R_{S'_e w'}$	= correlation coefficient between fluctuations in supersaturation and z-velocity [-]
ρ_a, ρ_l, S_e	= density of air [kg/m ³], density of water [kg/m ³], environmental supersaturation [-]
S_k, S_c	= particle equilibrium supersaturation [-], critical equilibrium supersaturation [-]
S_t, σ_l	= Stokes number [-], surface tension of water [N/m]
T, τ_p	= instantaneous temperature [K], particle response time [s]
$\mathbf{u}$	= instantaneous fluid velocity vector [m/s]
$\mathbf{V}, \mathbf{X}$	= particle velocity vector [m/s], particle position vector [m]
PSD	= particle size distribution [$\mu\text{m}^{-1}\text{cm}^{-3}$]
PDF, TKE	= probability density function [1/unit], turbulent kinetic energy [m ² /s ³]

I. INTRODUCTION

Numerical modeling has become a powerful tool to investigate turbulence-cloud-aerosol interactions in the Earth's atmosphere as experiments and measurements are difficult and expensive to conduct. The most accurate approach is direct numerical simulation (DNS) coupled with Lagrangian particle tracking (Liu et al.¹). This method resolves the flow of turbulent air up to the smallest scales without any kind of parameterizations of small turbulent eddies. It also tracks cloud droplets and aerosol particles individually. Two thermodynamic (scalar) fields, namely temperature and water vapor mixing ratio, are transported in the Eulerian framework. Flow turbulence is assumed to be homogeneous and isotropic, but it decays in the absence of a turbulence production mechanism. Many DNS studies (Vaillancourt,² Kumar et al.,³ Li⁴) and our previous DNS studies (Gao et al.,⁵ Sharfuddin et al.⁶) have applied external forcing in the momentum field to maintain turbulence. However, thermodynamic fluctuations decayed in those studies. In the atmosphere, thermodynamic fluctuations are sustained by mechanisms such as the entrainment of ambient air into the cloud and solar radiation absorbed by the cloud.^{7,8,9} Numerical representation of these events requires the addition of external forcing in the thermodynamic fields.

The idea of forcing is to inject energy to a band of low wavenumbers from they are transferred to the high wavenumber region of the turbulence spectrum. Eswaran and Pope¹⁰ showed that Fourier forcing in the momentum field results in a statistically stationary velocity field. Bogucki et al.¹¹, Watanabe and Gotoh¹², and Chen and Cao¹³ applied Fourier forcing in a passive scalar (thermodynamic) field. They suggest that the scaling behavior of a passive scalar differs from that of the velocity field because scalar fluctuations are highly intermittent at small scales and become decorrelated more rapidly than velocity fluctuations do. In the context of atmospheric clouds, the DNS model of Paoli and Shariff¹⁴ implemented Fourier forcing in the momentum, temperature, and water vapor fields. The authors reported that the temperature and water vapor fields reach statistical stationarity due to the forcing. But their model's setup did not account for latent heat release and turbulent entrainment-mixing processes. The DNS model of Chen et al.¹⁵ forced the thermodynamic fields, also in a low wavenumber band, to simulate the growth of mixed-phase clouds. Satio et al.¹⁶ applied Fourier forcing in the water vapor field but kept the temperature field unforced.

Thermodynamic forcing influences cloud microphysics in numerical models. Paoli and Shariff¹⁴ suggested that the temperature and vapor fluctuations sustained by forcing broaden the size distributions of cloud droplets. Satio et al.¹⁶ found that droplet size distributions are narrower in their DNS model relative to those observed in the experiments if thermodynamic forcing is absent, but that there is better agreement with forcing. The DNS model of Thomas et al.¹⁷,

in which the adiabatic lapse rate term acted as temperature forcing and the water vapor field was unforced, showed that supersaturation fluctuations depend on temperature forcing and affect the condensation of cloud droplets. Using a similar DNS model, Satio and Gotoh¹⁸ reported that the microphysical fluctuations are strongly impacted by large-scale turbulence. MacMillan et al.¹⁹ conducted a DNS investigation of a laboratory cloud chamber, with the sidewalls acting as a source of forcing in the water vapor field. They found that the probability distributions of supersaturation are non-Gaussian and the size distributions of cloud droplets broaden or narrow depending on the details of simulation parameters.

Despite the abovementioned studies, knowledge gaps still exist regarding the role of thermodynamic forcing in microscale cloud models. Specifically, the interaction between latent heat release or absorption and thermodynamic forcing requires better understanding. Also, microphysical events like condensation, aerosol activation, evaporation and cloud droplet deactivation need to be understood under sustained thermodynamic fluctuations. In this paper, we extend our previous DNS study⁶ to include external forcing in the temperature and vapor mixing ratio fields. The forcing is carried out at the low wavenumbers using the approach of Fourier forcing.¹⁰ Note that thermodynamic forcing is added in addition to the forcing of the base momentum field. The inter-conversion of aerosol particles and cloud droplets is modeled using the kappa-Kohler theory.²⁰ Simulations are conducted in three spatial dimensions and time for two sets of initial conditions. The forced and unforced thermodynamic fields are compared in terms of the mean, variance, energy spectrum, correlation, and probability distribution. The effects of varying the magnitudes of the critical thermodynamic and momentum forcing metrics are investigated. The microscale processes of condensation and evaporation are studied along with different regimes of activation and deactivation. The impact of external forcing on the broadening and narrowing of particle size distributions is also examined for underlying physical mechanisms.

II. GOVERNING EQUATIONS

The equations solved in our model are described in this section. They include the Navier-Stokes equations for the flow field, the conservation equations for the thermodynamic variables, the models for the Lagrangian particle tracking, microphysical equations, and the kappa-Kohler theory which models the curvature and solute effects.

A. The velocity field

The turbulent flow of ambient air is assumed to be governed by the incompressible Navier-Stokes equations with the Boussinesq approximation:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_a} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F}_b + \mathbf{F}_c, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Here, $\mathbf{u} = (u, v, w)$ is the instantaneous fluid (air) velocity vector, p is the instantaneous fluid static pressure, and $\mathbf{F}_b$ is the buoyancy force which is implemented in the form:²¹

$$\mathbf{F}_b = -\mathbf{g} \left[\frac{T - \langle T \rangle}{\langle T \rangle} + 0.608(q_v - \langle q_v \rangle) - q_c \right]. \quad (3)$$

Where $\mathbf{g}$ is the acceleration vector due to normal gravity, T is the instantaneous temperature, q_v is the instantaneous water vapor mixing ratio and q_c is the liquid water mixing ratio. The $\langle T \rangle$ and $\langle q_v \rangle$ are the domain averages of T and q_v , respectively. Our model's domain size (0.512 m) is much smaller compared to an actual cloud and thus $\mathbf{F}_b$ is found to be relatively insignificant and unable to maintain flow turbulence via buoyancy production. As a result, the momentum field is forced by adding the $\mathbf{F}_c$ in Eq. (1). The $\mathbf{F}_c$ is formulated in the Fourier space as:²²

$$\hat{\mathbf{f}}_c(\mathbf{k}, t) = \epsilon_{in} \frac{\hat{\mathbf{u}}(\mathbf{k}, t)}{\sum_{\mathbf{k}_f \in k} |\hat{\mathbf{u}}(\mathbf{k}_f, t)|^2} \delta_{\mathbf{k}, \mathbf{k}_f}. \quad (4)$$

where ϵ_{in} is an input turbulent kinetic energy (TKE) dissipation rate, $\mathbf{k}$ is the wavenumber vector, $\hat{\mathbf{u}}(\mathbf{k}, t)$ is the velocity field in the Fourier space, $\mathbf{k}_f$ is the forced wavenumber vector, and $\delta_{\mathbf{k}, \mathbf{k}_f}$ is a delta function which is unity inside the forced wavenumber shell but zero otherwise. The $\hat{\mathbf{f}}_c$ is inverse-transformed into the physical space as $\mathbf{F}_c$. Note that the “back and forth” with the Fourier and physical space is necessary because our numerical procedure takes place in the physical space, while the subset of the spectrum that is forced can only be directly obtained in Fourier space.

B. Conservation of the thermodynamic fields

The conservation equations for temperature (T) and water vapor mixing ratio (q_v) are solved in the following form:

$$\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T = \frac{L_h}{c_p} C_d + \mu_T \nabla^2 T + F_T, \quad (5)$$

$$\frac{\partial q_v}{\partial t} + (\mathbf{u} \cdot \nabla) q_v = -C_d + \mu_v \nabla^2 q_v + F_{q_v}. \quad (6)$$

where C_d quantifies the time rate of exchange between liquid water and water vapor and is described by Eq. (17). The phase change terms in Eqs. (5) and (6) are $L_h C_d / c_p$ and $-C_d$, respectively, where L_h is the specific latent heat and c_p

is the specific heat of air. The molecular diffusivities of temperature and water vapor are μ_T and μ_v , respectively, while the F_T and F_{qv} are the external forcing terms in these equations. Analogous with momentum forcing, the forcing of a generic fluctuating thermodynamic field θ , F_θ , can be written in the Fourier space as:

$$\hat{f}_\theta(\mathbf{k}, t) = \epsilon_{\theta, in} \frac{\hat{\theta}(\mathbf{k}, t)}{\sum_{\mathbf{k}_f \in \mathbf{k}} |\hat{\theta}(\mathbf{k}_f, t)|^2} \delta_{\mathbf{k}, \mathbf{k}_f}, \quad (7)$$

where $\hat{\theta}(\mathbf{k}, t)$ is the fluctuating thermodynamic field in the Fourier space and $\epsilon_{\theta, in}$ is an input dissipation rate for θ . The $\hat{f}_\theta(\mathbf{k}, t)$ is converted back to the physical space as $F_\theta(\mathbf{x}, t)$ after selecting the wavenumbers that are forced. The domain-average of F_θ is zero. The external forcing terms in the momentum and thermodynamic fields are randomly generated and are independent of each other. Note that the impact of adiabatic lapse rate on temperature change is also examined and is found to be negligible (see Appendix A for discussion).

Dynamics of cloud and aerosol particles

The motion of each aerosol particle or cloud droplet is modeled in the Lagrangian fashion:³

$$\frac{d\mathbf{X}_i(t)}{dt} = \mathbf{V}_i(t), \quad (8)$$

$$\frac{d\mathbf{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i} [\mathbf{u}_i(\mathbf{X}, t) - \mathbf{V}_i(t)] + \mathbf{g}, \quad (9)$$

where the subscript ' i ' denotes the i -th aerosol particle or cloud droplet, $\mathbf{V}$ is its velocity vector, and $\mathbf{X}$ is the position vector. The $\mathbf{u}_i$ is the instantaneous fluid velocity vector at the position of particle i and is obtained through a trilinear interpolation of the Eulerian field. The particle response time τ_p is given by³

$$(\tau_p)_i = \frac{2\rho_l r_i^2}{9\rho_a \nu}. \quad (10)$$

where ρ_l is the density of liquid water, ρ_a is the air density, and ν is the kinematic viscosity of air. Equation (9) is valid when the particle Stokes number (S_t) is significantly less than unity. The time evolution of particle radius r is described by²³

$$\frac{dr_i(t)}{dt} = \frac{G_i}{r_i(t)} (S_e(\mathbf{X}, t) - S_k(\mathbf{X}, t))_i. \quad (11)$$

where S_e is the environmental supersaturation and S_k is the particle equilibrium supersaturation that accounts for the combined curvature and solute effects. The kappa-Kohler theory²⁰ expresses the S_k as:

$$S_k = \frac{A}{r} - \frac{\kappa r_d^3}{r^3 - r_d^3(1 - \kappa)}, \quad A = \frac{2\sigma_l M_l}{RT\rho_l}. \quad (12)$$

where r_d is the dry aerosol radius, κ is the hygroscopicity of solute (dry aerosol), R is the universal gas constant, σ_l is the surface tension of water, and M_l is molar mass of water. If the atmosphere is supersaturated, aerosol particles grow beyond a threshold, known as the critical radius, and activate into cloud droplets. Note that a particle experiences non-equilibrium, diffusion-limited growth once it passes the critical barrier.²⁴ In a subsaturated environment, cloud droplets shrink below the critical radius and deactivate into aerosol particles. Using the kappa-Kohler theory, the critical radius r_c and critical equilibrium supersaturation S_c are derived as:²⁵

$$r_c = \sqrt{\frac{3\kappa r_d^3}{A}}, \quad S_c = \frac{2}{\sqrt{\kappa}} \left(\frac{A}{3r_d} \right)^{\frac{3}{2}}. \quad (13)$$

A particle can be classified as aerosol or cloud by comparing its wet radius (r) with r_c . An aerosol particle corresponds to $r_d < r \leq r_c$ while a cloud droplet corresponds to $r > r_c$. The variable S_e depends on the variables q_v and T and is expressed as:

$$S_e = \frac{q_v}{q_{v,s}} - 1, \quad q_{v,s} = 621.97 \frac{p_{sat}}{p - p_{sat}}, \quad (14)$$

$$p_{sat} = 611.2 \times \exp\left(\frac{17.67(T - 273.15)}{T - 29.65}\right). \quad (15)$$

$q_{v,s}$ is the saturation vapor mixing ratio²⁶ and p_{sat} is the saturation vapor pressure.²⁷ The growth parameter G in Eq. (11) is found as:²³

$$G = \frac{1}{\frac{L_h \rho_l}{k_T T} \left(\frac{L_h}{R_v T} - 1 \right) (1 + S_k) + \frac{\rho_l R_v T}{\mu_v p_{sat}}}, \quad (16)$$

where k_T is the thermal conductivity in air and R_v is the specific gas constant for water vapor. The C_d is determined as:⁶

$$C_d(\mathbf{x}, t) = \frac{4\pi\rho_l}{\rho_a a^3} \sum_{i=1}^n G_i r_i(t) (S_e(\mathbf{X}, t) - S_k(\mathbf{X}, t))_i. \quad (17)$$

where n is the number of particles in a grid cell that has a volume of a^3 . In the present study, the value of n is 0.95 and that of a is 0.512 m.

III. NUMERICAL DETAILS

We run simulations for two scenarios: Scenario I and Scenario II.⁶ Since the domain size of our DNS model is much smaller than that of a natural cloud, it will be difficult to observe significant condensation and evaporation at the same time within our model volume. As a result, we designed two separate scenarios that correspond to different initial conditions. In the first scenario, the cloud volume mixes with saturated environmental air, and activation and condensation prevail. In the second, the dry (subsaturated) air is entrained into the cloud volume and evaporation and deactivation are dominant.

A. Momentum, thermodynamic, and particle fields

To examine the influences of turbulence intensity, we vary the range of forced wavenumbers to generate two levels of flow turbulence: ‘low’ and ‘high,’ and we denote them as ‘L’ and ‘H,’ respectively.⁶ The forcing wavenumber bands are: $1 \leq k_f/k_{min} \leq 4.12$ and $1 \leq k_f/k_{min} \leq 16.18$, for the two levels of turbulence. The minimum wavenumber $k_{min} = 2\pi/L = 2\pi/0.512 \sim 12.27$ and the maximum wavenumber $k_{max} \sim 128k_{min}$. The initial TKE spectrum is specified as²⁸

$$E(k) = \frac{16}{\sqrt{\pi/2}} \frac{u_0^2 k^4}{k_0^5} \exp\left(-\frac{2k^2}{k_0^2}\right). \quad (18)$$

where $k_0 = 4.12k_{min}$ and $u_0 = 0.16 \text{ m/s}$ is the root-mean-square (rms) velocity from the generated fluctuating velocity field. The form k^4 for low wavenumbers correspond to the incompressible limit while the exponential function allows a rapid roll-off of the energy at high wavenumbers. Table I lists the statistics of the statistically stationary turbulence field. The data includes the TKE dissipation rate, ϵ , the Taylor microscale, λ , the Kolmogorov length scale η , the length scale characterizing large eddies, l_0 , the forcing Reynolds number, Re_f , the Taylor-scale Reynolds number, Re_λ , and the Reynolds number based on l_0 , Re_{l_0} . The definition of abovementioned quantities can be found in Tennekes and Lumley.²⁹ The values in Table I match the parameters of ambient turbulence as measured by Siebert et al.³⁰

Table I. Statistics of the turbulent velocity field at 4 s

Turbulence level	ϵ (m^2/s^3)	λ (m)	η (m)	l_0 (m)	Re_f	Re_λ	Re_{l_0}
High	0.00723	0.0169	0.00082	0.184	455.41	156.11	1704
Low	0.00063	0.0261	0.00153	0.227	201.90	138.74	1207

Our model is that of a 3D cube inside which flow turbulence, cloud droplets, and aerosol particles interact. The boundary conditions are triply periodic, following the standard procedure of DNS. The domain volume is 0.512^3 m^3 . At the start of simulations, the domain is divided equally into cloudy and clear-air regions. A slab-like configuration⁴ is assumed for the initial field (Fig. 1) to enable a sharp transition in the water vapor mixing ratio and temperature at the cloud/clear-air interface. The initial thermodynamic fields change abruptly across the interface in our model unlike those in Paoli and Shariff¹⁴ and Chen et al.¹⁵ Such an inhomogeneous distribution of the initial thermodynamic fields allows us to realistically simulate turbulent mixing between the cloud and environment.³¹ However, this configuration precludes the use of pseudospectral method, which otherwise gives the Gibbs phenomenon at the interfaces² which is unphysical. This partially explains the use of the finite difference method¹⁹ in the current study.

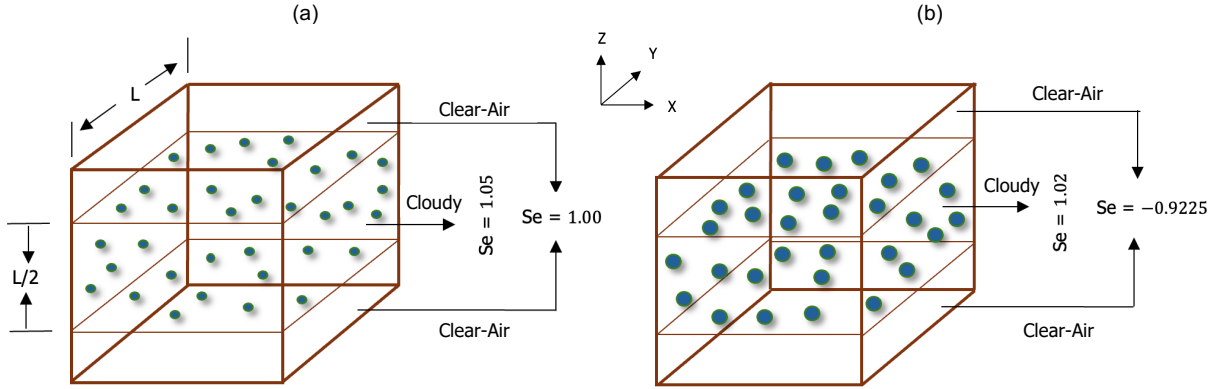


FIG. 1 Initial configuration in (a) Scenario I and (b) Scenario II. The middle half represents the cloudy region while the top and bottom one-fourths constitute the clear-air region. A sharp interface separates the two regions. The cloudy region is initially 5% supersaturated ($S_e = 1.05$) in Scenario I and 2% ($S_e = 1.02$) in Scenario II. The clear-air region is just saturated in the first ($S_e = 1.00$) and 92.25% subsaturated ($S_e = -0.9225$) in the second. The spheres represent dry aerosols in Scenario I and cloud droplets in Scenario II.

In Scenario I, at the initial time, the cloudy region is supersaturated ($S_e = 1.05$), while the clear-air region is just saturated ($S_e = 1.00$). To maintain the distinction between cloud and clear air, the initial vapor mixing ratio field is specified as:⁵

$$q_v(z, t = 0) = \begin{cases} q_v^{max} = 1.05 \times q_{v,s}, & d - L \times 0.5 \times 0.5 \leq z < d + L \times 0.5 \times 0.5 \\ q_{v,e} = q_{v,s}, & \text{elsewhere} \end{cases}. \quad (19)$$

Above, $q_{v,e} = q_{v,s} = 3.872 \text{ g/kg}$ represent the clear-air region and $q_v^{max} = 4.066 \text{ g/kg}$ signify the cloudy region.

The length, $d \equiv L/2$ is the width of the cloud slab. The temperature field is initialized as:⁵

$$T(t = 0) = \langle T(t = 0) \rangle - 0.608 \langle T(t = 0) \rangle (q_v(z, t = 0) - \langle q_v(t = 0) \rangle). \quad (20)$$

Equation (20) uses the information of initial water vapor field and assigns one temperature value to the cloudy region while another to the clear-air region. Based on Eqs. (19) and (20), the mean temperature and mean water vapor mixing

ratio are not zero unlike the mean velocity. A Gaussian random field is superimposed on the initial temperature and water vapor fields, and it follows the spectrum:

$$E_\theta(k) = \frac{16}{\sqrt{\pi/2}} \frac{\theta_0^2 k^4}{k_0^5} \exp\left(-\frac{2k^2}{k_0^2}\right). \quad (21)$$

where $\theta_0 = \langle T(t=0) \rangle \times 10^{-4}$ for the temperature field while $\theta_0 = \langle q_v(t=0) \rangle \times 10^{-3}$ for the water vapor field.³² Again, $k_0 = 4.12k_{min}$. We simulate five cases for Scenario I, as summarized in Table II. These cases are generated by varying the level of momentum and thermodynamic forcing. The forcing wavenumber bands corresponding to ‘low’ and ‘high’ levels of thermodynamic forcing are: $1 \leq k_f/k_{min} \leq 4.12$ and $1 \leq k_f/k_{min} \leq 16.18$, respectively. At the start of simulations, 16×10^6 dry aerosols are randomly placed in the cloudy region which gives a number concentration of 119 cm^{-3} . The size (radius) distribution of dry aerosols is assumed to be monodisperse with a mean of $0.1 \text{ }\mu\text{m}$ for the five cases in Table II. The assumed dry aerosol parameters (119 cm^{-3} and $0.1 \text{ }\mu\text{m}$) are typical in stratocumulus clouds.³³ Dry aerosols absorb water from the environment to become aqueous aerosol particles which grow further and activate into cloud droplets in Scenario I.

TABLE II. Test cases for Scenario I and Scenario II.

Case	TKE dissipation rate (m^2/s^3)	Level of momentum forcing (F_C)	Level of thermodynamic forcing (F_θ)	Initial particle size (μm) in Scenario I	Initial particle size (μm) in Scenario II
L	0.00063	Low	N/A	0.1	15
L-L	0.00063	Low	Low	0.1	15
L-H	0.00063	Low	High	0.1	15
H	0.00723	High	N/A	0.1	15
H-L	0.00723	High	Low	0.1	15
Case	Comments				
L	Level of momentum forcing is low (L). Thermodynamic forcing is absent or $F_T = F_{qv} = 0$.				
L-L	Level of momentum forcing is low (L). Level of thermodynamic forcing is low (L).				
L-H	Level of momentum forcing is low (L). Level of thermodynamic forcing is high (H).				
H	Level of momentum forcing is high (H). Thermodynamic forcing is absent or $F_T = F_{qv} = 0$.				
H-L	Level of momentum forcing is high (H). Level of thermodynamic forcing is low (L).				

In Scenario II, the cloudy region is supersaturated ($S_e = 1.02$) but the clear-air region is subsaturated ($S_e = -0.9225$) at the initial time. The two regions correspond to $q_v^{max} = 3.949 \text{ g/kg}$ and $q_{v,e} = 0.30 \text{ g/kg}$, respectively. A total of 16×10^6 cloud droplets, each of them having a radius r of $15 \text{ }\mu\text{m}$, are randomly placed in the cloudy region

at the initial time. Dry aerosols, also known as solutes, are assumed to be present in the core of cloud droplets with $r_d = 0.1 \mu m$. For a known r_d , the r_c can be calculated using Eq. (13). The r_d and r_c are fixed at each particle position while the wet radii r evolves with time. We again simulate five cases (Table II) in Scenario II which correspond to evaporation and deactivation of cloud droplets.

B. Parameters, grid resolution, and computing resources

Table III lists the parameters used in our model including the domain-averaged temperature, pressure, and Stokes number (S_t) at initial time. The chemical composition of dry aerosol is ammonium sulfate with a hygroscopicity (κ) value of 0.61.³⁴ Particles are allowed to grow (Eq. (11)) after the TKE dissipation rate becomes statistically stationary which takes 2 s. The input dissipation rates in the temperature ($\epsilon_{T,in}$) and vapor mixing ratio ($\epsilon_{qv,in}$) fields and the Stokes numbers at initial time are shown in Table III. Note that the mean temperature, pressure, and density do not vary with height in our microscale model. A detailed analysis in this regard can be found in Ref. 6.

TABLE III. Parameters

Symbol	Value	Unit	Symbol	Value	Unit
q_v^{max}	4.066 (Scenario I)	g/kg	$q_{v,e}$	3.872 (Scenario I)	g/kg
	3.949 (Scenario II)			0.30 (Scenario II)	
L_h	2.5×10^6	J/kg	c_p	1005.0	J/kg/K
ρ_a	0.92	kg/m ³	ν	1.5×10^{-5}	m ² /s
ρ_l	1000	kg/m ³	$\langle p(t = 0) \rangle$	71446	N/m ²
$\langle T(t = 0) \rangle$	270.75	K	R_v	461.5	J/kg/K
k_T	0.0238	W/m/K	σ_l	0.072	N/m
$\mu_T = \mu_v$	2.2×10^{-5}	m ² /s	M_l	0.018	kg/mol
R	8.314	J/mol/K	$\langle S_t(t = 0) \rangle$	9.6×10^{-7} (Scenario I)	-
κ	0.61	-		2.16×10^{-2} (Scenario II)	
$\epsilon_{T,in}$	1×10^{-5} (Scenario I)	K ² /s	$\epsilon_{qv,in}$	1×10^{-4} (Scenario I)	(g/kg) ² /s
	1×10^{-3} (Scenario II)			1×10^{-2} (Scenario II)	

As a compromise between grid resolution and computational cost, we use **256³** grid points for all simulations. The $k_{max}\eta$ is a dimensionless parameter that determines the sufficiency of grid resolution in DNS and is prescribed to be at least greater than 1.³⁵ The $k_{max}\eta$ for the ‘low’ and ‘high’ turbulence levels are 2.46 and 1.43, respectively. The details of the numerical solvers are available in Ref. 6. The Perlmutter supercomputer, managed by the National

Energy Research Scientific Computing Center (NERSC), United States Department of Energy, USA, has been used for the simulations.

IV. RESULTS AND DISCUSSIONS

We discuss the impact of external forcing on the fluctuating and mean thermodynamic fields in Sec. IV A. The impacts of thermodynamic forcing on condensation, aerosol activation, evaporation, and cloud droplet deactivation are explained in Sec. IV B and Sec. IV C. The concluding remarks in Sec. V summarize our findings.

A. The impact of external forcing on the thermodynamic fields

Thermodynamic and momentum forcing influences the evolution of mean thermodynamic fields in our model as shown in Figs. 2(a)-(f). The mean temperature $\langle T \rangle$ increases during condensation (Fig. 2(a)) but decreases during evaporation (Fig. 2(d)) if thermodynamic forcing is absent (Cases L and H) due to condensational heating and evaporative cooling which take place in clouds.³⁶ The mean vapor mixing ratio $\langle q_v \rangle$ decreases in Scenario I (Fig. 2(b)) but increases in Scenario II (Fig. 2(e)) for Cases L and H due to the fact that water vapor turns to liquid water during condensation and liquid water becomes water vapor during evaporation. The change of mean supersaturation $\langle S_e \rangle$ in Figs. 2(c) and 2(f) resembles that of $\langle q_v \rangle$ since $\langle S_e \rangle$ is linearly proportional to $\langle q_v \rangle$. In the absence of thermodynamic forcing, the domain becomes saturated ($\langle S_e \rangle \sim 0$) in both scenarios, even though their initial conditions differ. The difference between thermodynamically forced and unforced cases becomes prominent after 20 s in Scenario I and after 15 s in Scenario II. The changes of $\langle T \rangle$, $\langle q_v \rangle$, and $\langle S_e \rangle$ become asymptotic over time for the unforced cases. On the other hand, the temporal increase of $\langle T \rangle$ in Scenario I (Fig. 2(a)) is followed by a decrease if temperature and vapor fields are forced. The decrease conforms to the order: L-H > H-L > L-L, implying that a higher level of thermodynamic forcing (Case L-H) enhances the decay of $\langle T \rangle$. The $\langle q_v \rangle$ decays similarly in Scenario I (Fig. 2(b)) and in Scenario II (Fig. 2(e)) for three forced cases. If thermodynamic forcing is absent (L and H), momentum forcing affects the $\langle T \rangle$, $\langle q_v \rangle$, and $\langle S_e \rangle$ until saturation ($\langle S_e \rangle \sim 0$) is reached and not afterwards. Strong turbulence in Case H enhances mixing between the cloud and dry air, leading to a higher evaporation rate and faster increase of $\langle q_v \rangle$ compared to Case L (Fig. 2(e)). However, strong mixing reduces cloud supersaturation, suppresses condensation, and slows down the decrease of $\langle q_v \rangle$ for Case H relative to Case L (Fig. 2(b)).

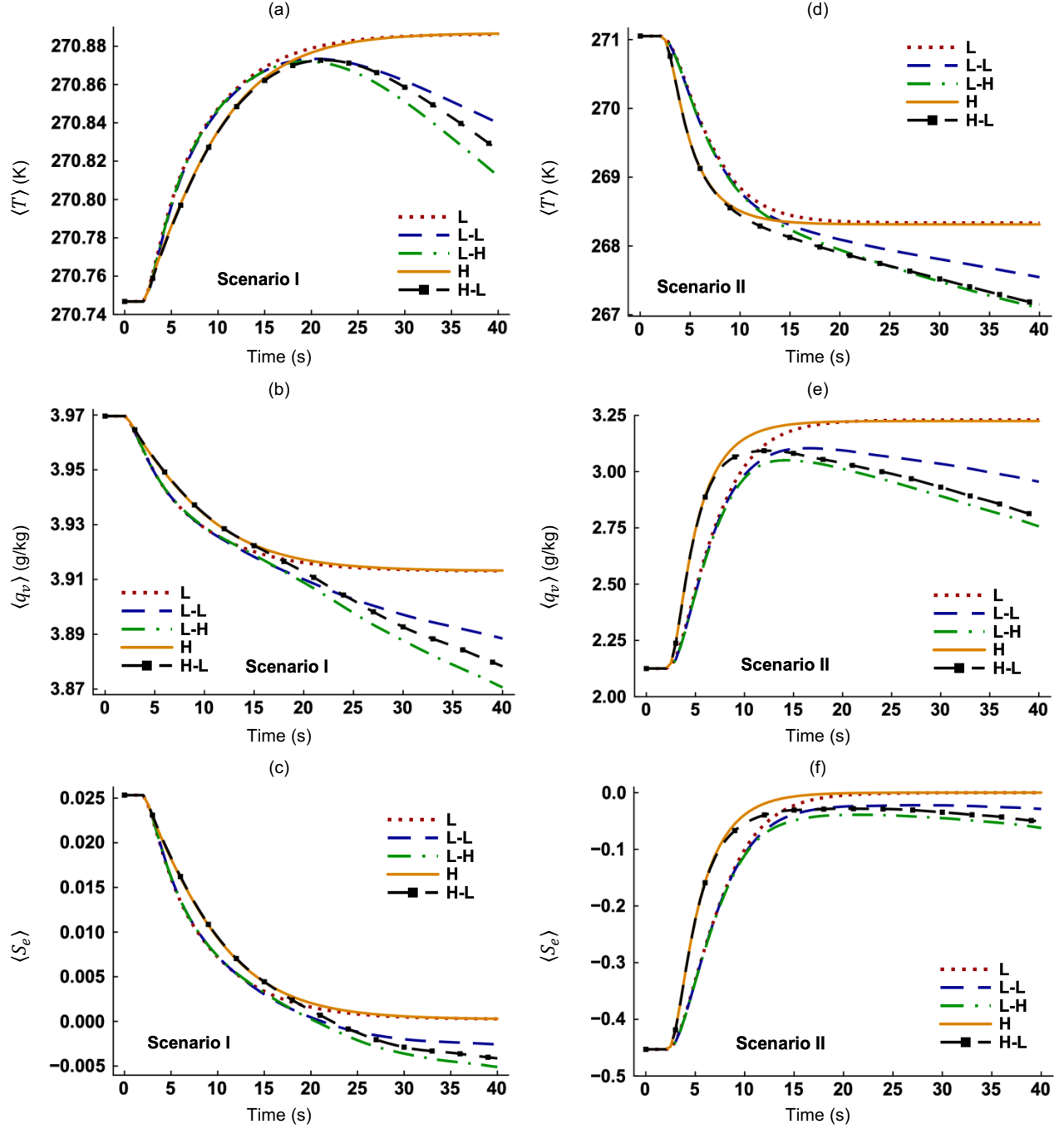


FIG. 2 Time evolution of the (a) mean temperature ($\langle T \rangle$), (b) mean vapor mixing ratio ($\langle q_v \rangle$), and (c) mean environmental supersaturation ($\langle S_e \rangle$), respectively in Scenario I. Time evolution of the (d) mean temperature ($\langle T \rangle$), (e) mean vapor mixing ratio ($\langle q_v \rangle$), and (f) mean environmental supersaturation ($\langle S_e \rangle$), respectively in Scenario II.

The temporal evolution of the variances of temperature (σ_T^2) and vapor mixing ratio (σ_{qv}^2) is shown in Figs. 3(a)-(d). We see that thermodynamic fluctuations attain statistical stationarity when thermodynamic forcing is applied (Cases L-L, L-H, and H-L), but they decay otherwise (Cases L and H). A higher level of momentum forcing enhances mixing and phase change, resulting in a faster decay of variances for Case H compared to Case L (Figs. 3(a)-(d)). On the other hand, thermodynamic forcing acts as a source of fluctuations and counteracts the decay (Cases L-L, L-H, and H-L). The magnitude of temperature fluctuations is larger in Scenario II (Fig. 3(c)) compared to Scenario I (Fig. 3(a)) and the same is true for water vapor fluctuations (Fig. 3(d) vs Fig. 3(b)) because the initial conditions for the thermodynamic and particle fields are different in two scenarios (Sec. III A).

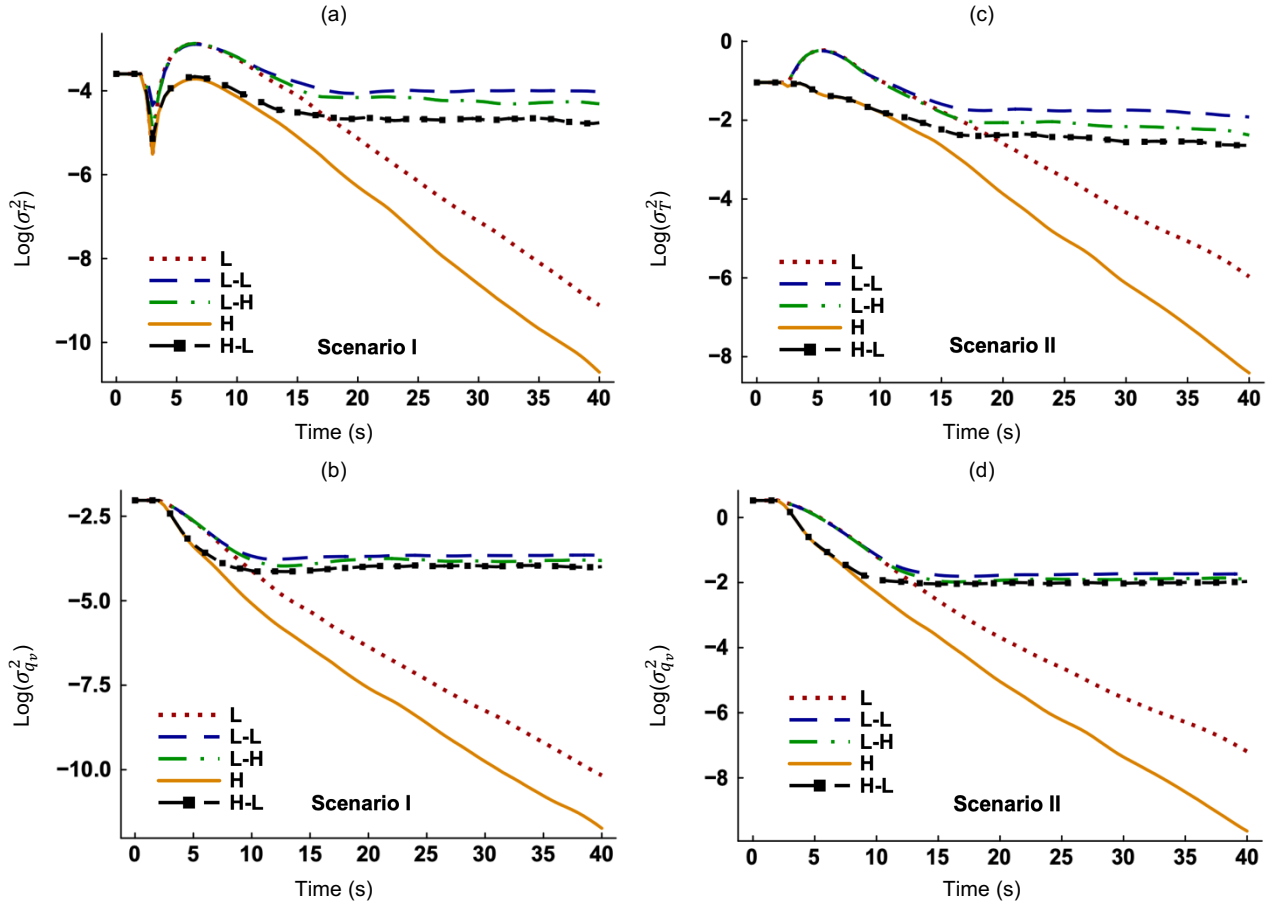


FIG. 3 Time evolution of the variances of (a) temperature and (b) water vapor mixing in Scenario I, respectively. Time evolution of the variances of (c) temperature and (d) water vapor mixing in Scenario II, respectively.

The energy spectra of temperature ($E_T(k)$) and vapor mixing ratio ($E_{qv}(k)$) are shown in Figs. 4(a)-(b) for the five cases at 20 s in Scenario I. For Cases L and H, only momentum forcing is present and the $E_T(k)$ and $E_{qv}(k)$ differ at large scales but converge at small scales. This is consistent with turbulence theory as kinetic energy injected at large scales breaks down into progressively smaller scales, leading to universal statistics at small scales.³⁵ The integrals of

the $E_T(k)$ and $E_{qv}(k)$ appear to be larger for Case L compared to Case H at 20 s, implying more energy for the former and consistent with Fig. 3. Interestingly, the spectra briefly flatten at around $\log(k) \sim 3.0$ or $k \sim 1000$ (Figs. 4(a)-(b)) before the final decay if thermodynamic forcing is absent (L and H). This is due to energy injection at small scales by the phase change of particles. For the thermodynamically forced cases (L-L, L-H, and H-L), the interaction between external forcing (F_θ) and phase change (C_d) increases energy at all scales (Figs. 4(a)-(b)) even though F_θ is applied at large scales only. The slope of the spectra in the inertial subrange quantifies their decay. We find that the slope for Cases H and H-L in Fig. 4(a) is -2.9 while that for Cases L, L-L, and L-H is -4.25 . In Fig. 4(b), the slope for Cases L and H-L is -3.0 and that for Case H is -2.6 while that for Cases L-L and L-H is -3.4 . These values suggest that the decay of thermodynamic spectra is faster than that of Kolmogorov's spectra ($-5/3 = -1.67$) in homogeneous and isotropic turbulence (HIT). Similar observation was made by Ladeinde and Oh³⁷ who studied the transport of a passive scalar in compressible HIT. They reported that the slope of spectral decay is -4 in the scalar field.

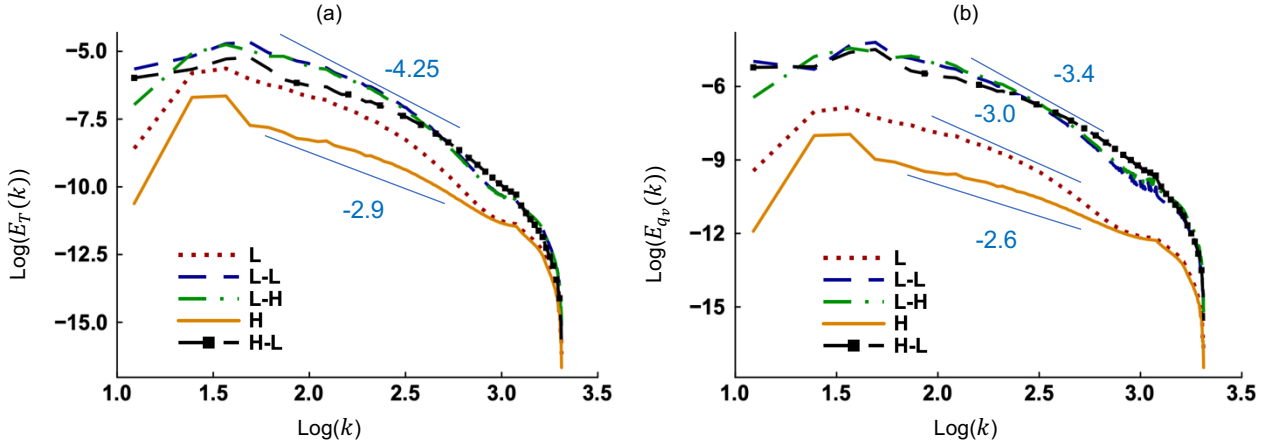


FIG. 4 Energy spectra of the (a) temperature and (b) water vapor mixing ratio for the five cases. Both the ordinate and abscissa are on a base-10 logarithmic scale. These results correspond to Scenario I at 20 s. The spectral profiles are similar for the other datasets including Scenario II. The spectra of S_e resemble that of q_v and hence are not shown. The wavenumbers (k) are multiples of k_{min} and are in the radial coordinate direction. The slopes of the curves in the inertial subrange are also shown.

We further investigate the impact of phase change on the $E_T(k)$ and $E_{qv}(k)$ in Figs. 5(a)-(b). Two cases: L and LA are compared. Thermodynamic forcing is absent for both cases while phase change is absent ($C_d = 0$) in the latter. We see in Figs. 5(a)-(b) that a flattening is observed at $\log(k) \sim 3.0$ only if phase change takes place (Case L) but not otherwise (Case LA). Moreover, the presence of C_d changes the magnitudes of $E_T(k)$ and $E_{qv}(k)$ at all scales. Note that phase change is a small-scale phenomenon and it involves the change of instantaneous particle radius (r). Hence, the energy spectra of r ($E_r(k)$) is shown in Fig. 5(c). We notice that the $E_r(k)$ is relatively large at the small scales

but negligible at larger scales and has a slope of 2.22 in the inertial subrange. This implies that an inverse energy cascade is present. Therefore, some molecular diffusion occurs at small scales, but with a negative diffusivity.

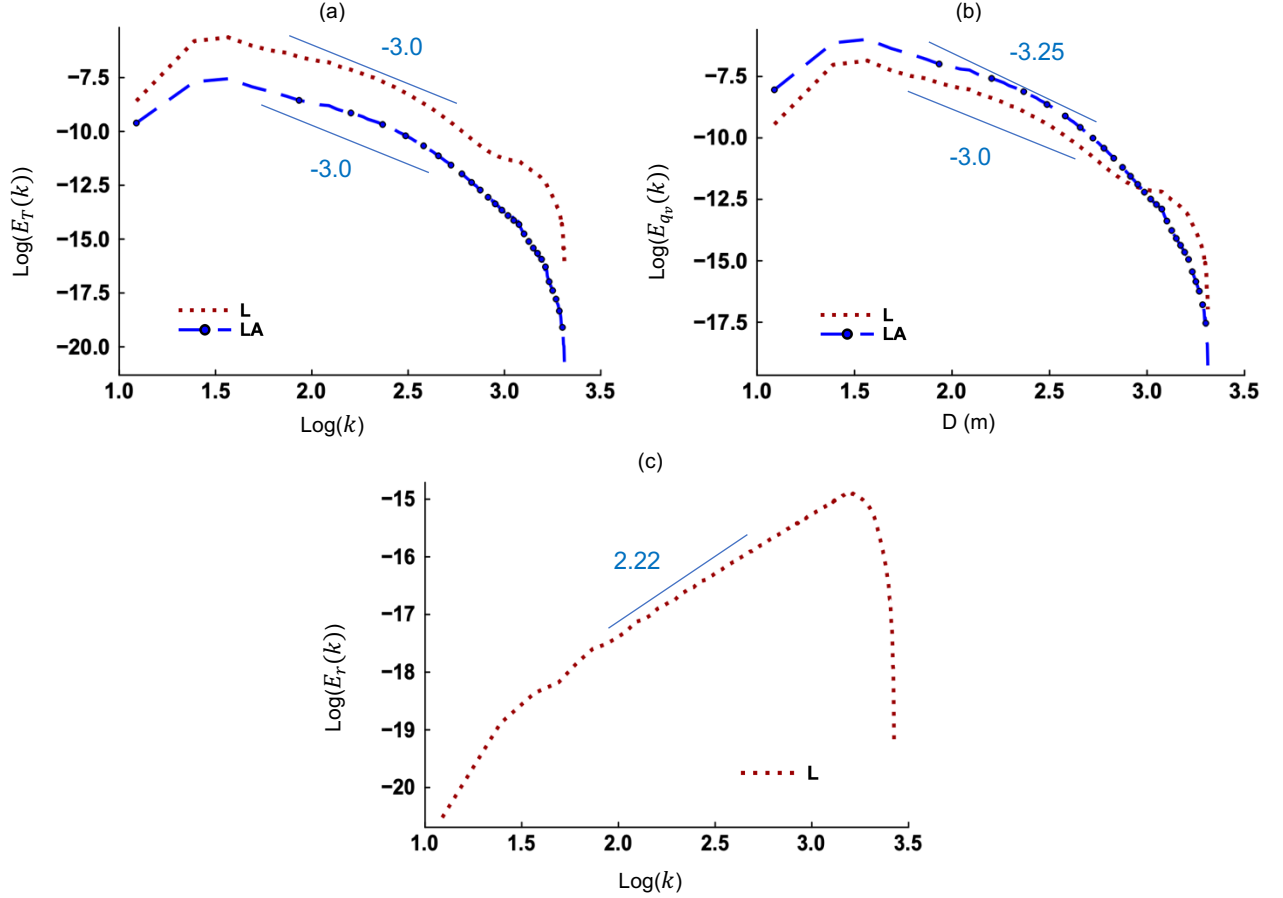


FIG. 5 Energy spectra of the (a) temperature and (b) vapor mixing ratio for Cases L and LA in Scenario I at 20 s. Energy spectrum of the (c) particle radii for Case L in Scenario I at 20 s. The slopes of the curves in the inertial subrange are also shown. Note that Case LA excludes phase change and has no $E_r(k)$.

To substantiate the idea of negative diffusivity, we solve a steady-state transport equation for water vapor by introducing the effective diffusivity:

$$(\mathbf{u} \cdot \nabla) q_v = -C_d + \mu_{v,eff} \nabla^2 q_v + F_{q_v}. \quad (22)$$

where $\mu_{v,eff}$ is the effective diffusivity for the q_v . Figure 6(a) shows that $\mu_{v,eff}$ varies in space. Whereas μ_v is assumed to be a constant ($2.2e - 5$). Interestingly, the $\mu_{v,eff}$ is negative at many points in the physical space, but the view in the wavenumber space (Fig. 6(b)) is more revealing. The $E_{\mu_{v,eff}}(k)$ confirms the backscatter or the transfer of energy from small to large scales³⁸ with a positive slope of 2.17. The inverse energy cascade, as suggested by our model, also exists in real atmosphere. Shao et al.³⁹ reported that the rapid rotation of a typhoon induces an inverse energy cascade

and strengthens the typhoon. Vallis et al.⁴⁰ found that microscale cooling or heating affects large-scale convective motion through upscale energy transfer.

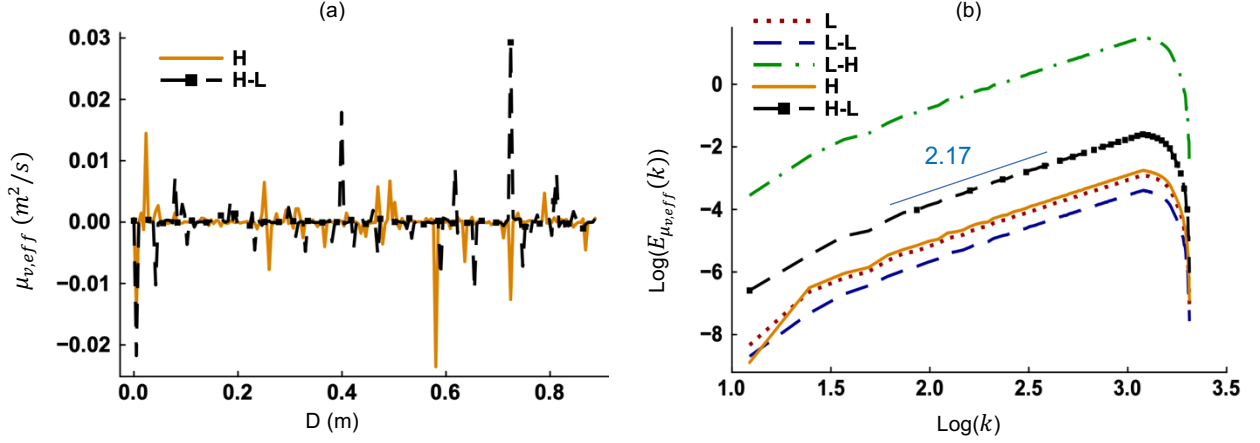


FIG. 6 (a) Spatial distribution of the effective vapor diffusivity for Cases L and L-H. (b) Energy spectra of the effective vapor diffusivity for the five cases. These results correspond to Scenario I at 25 s.

Fluctuations in the water vapor strongly influences fluctuations in the supersaturation if thermodynamic forcing is applied. The spatial correlation coefficient between fluctuations in the water vapor (q'_v) and supersaturation (S'_e) at a given time is defined as:³⁷

$$R_{q'_v S'_e}(\mathbf{r}) = \frac{\langle q'_v(\mathbf{x}, t) S'_e(\mathbf{x} + \mathbf{r}, t) \rangle}{\sqrt{\langle (q'_v(\mathbf{x}, t))^2 \rangle \langle (S'_e(\mathbf{x}, t))^2 \rangle}}, \quad \mathbf{r} = \Delta x \mathbf{i} + \Delta y \mathbf{j} + \Delta z \mathbf{k}. \quad (23)$$

where $\mathbf{r}$ is the separation distance vector. We see in Fig. 7(a) that q'_v and S'_e are strongly correlated at short separation distances for the forced cases (L-L, L-H, and H-L). The correlation is relatively weaker for the unforced cases (L and H). However, the correlation decays faster for the forced cases which have a correlation length scale of approximately 0.1 m. For all cases, the correlation decays as separation distance increases and significant memory no longer exists. The source term in Eq. (6) is the combination of external forcing and latent heat change. The fluctuating source term is $(F_{q_v} - C_d)' = F_{q_v} - C_d - \langle F_{q_v} - C_d \rangle$ and it exerts influence in the q'_v as shown in Fig. 7(b). Again, the $R_{q'_v (F_{q_v} - C_d)'}$ is stronger for three thermodynamically forced cases compared to two unforced cases. This is consistent with our finding (Fig. 3) that fluctuations are sustained when F_{q_v} is present. Figure 7(c) shows the cross-correlation between fluctuations in the supersaturation and vertical velocity (w') for Case L-L. Evidentially, S'_e and w' are not correlated, in agreement with the finding of Satio et al.¹⁶ This observation holds true for the other cases examined as well. The break of link between S'_e and w' is reportedly caused by entrainment and turbulent mixing processes.⁴¹

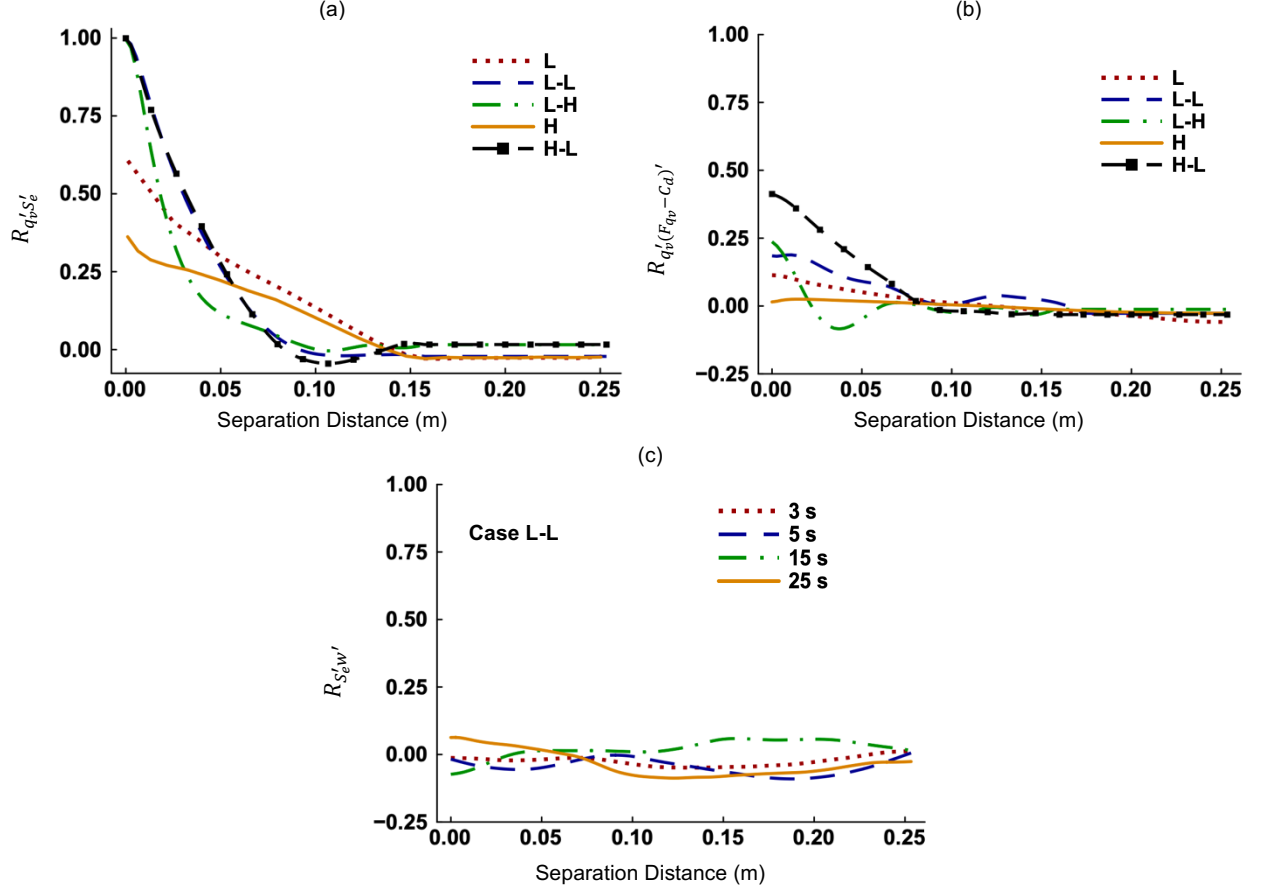


FIG. 7 (a) Correlation coefficient between fluctuations in the water vapor and supersaturation ($R_{q_v' S_e'}$). (b) Correlation coefficient between fluctuations in the q_v and $(F_{q_v} - C_d)$ which is denoted by $R_{q_v'(F_{q_v}-C_d)'}$. The plots represent the five cases in Scenario I at 25 s. (c) Correlation coefficient between fluctuations in the S_e and w ($R_{S_e' w'}$) for Case L-L in Scenario I at selected times.

We analyze the PDFs of thermodynamic fluctuations when they are statistically stationary. We take the data at 25 s in Scenario II as example. Figure 8(a) shows the PDFs of the fluctuating temperature normalized by the standard deviation (T'/σ_T) and Fig. 8(b) shows the same for water vapor mixing ratio. The observations in Figs. 8(a)-(b) are also valid for the other datasets. The PDFs of T'/σ_T and q_v'/σ_{q_v} do not appear to be Gaussian. To measure the deviation from Gaussianity, we calculate the skewness (γ) and kurtosis (ω) of the probability distributions as³⁷

$$\gamma = \frac{\langle(\varphi')^3\rangle}{\langle(\varphi')^2\rangle^{3/2}}, \quad \omega = \frac{\langle(\varphi')^4\rangle}{\langle(\varphi')^2\rangle^{4/2}}, \quad \varphi = \frac{\theta'}{\sigma_\theta}. \quad (24)$$

The γ for temperature PDFs in Fig. 8(a) are 0.25, -0.55, -0.66, 0.22, and -0.19 for Cases L, L-L, L-H, H, and H-L, respectively, while the corresponding ω are 2.94, 4.14, 3.48, 2.09, and 3.26. The γ for vapor PDFs in Fig. 8(b) are -0.22, -0.86, -0.82, 0.21, and -0.62, respectively, while the corresponding ω are 2.87, 4.56, 4.05, 2.09, and 3.80. Skewness is 0 and kurtosis is 3 for a perfectly Gaussian distribution. We see that both temperature and vapor PDFs

are negatively skewed for the forced cases (L-L, L-H, and H-L) while they are positively skewed for the unforced cases (L and H). Negative skewness implies that the distribution's tail extends toward smaller values and most values are larger than the mean. Thus, thermodynamic forcing enhances fluctuations at most nodal points. We find that the kurtosis of temperature and vapor PDFs are greater than 3 for the forced cases (L-L, L-H, and H-L) but less than 3 for the unforced cases (L and H). Consequently, the PDFs have a sharp peak with thermodynamic forcing (Figs. 8(a)-(b)) but have a rounded peak⁴² without forcing. Previous studies^{43,44} of cloud turbulence found the PDFs of thermodynamic fields to be non-Gaussian as this study shows. The mean thermodynamic gradients and microscale phase change cause intermittency and lead to non-Gaussian statistics. The magnitude of maximum T' in Figs. 8(a)-(b) is $\sim 4\sigma_T$ for Case L, $\sim 2\sigma_T$ for Case H, and $\sim 5\sigma_T$ for Cases L-L, L-H, and H-L. Again, the magnitude of maximum q'_v is $\sim 3\sigma_{q_v}$ for Case L, $\sim 2\sigma_{q_v}$ for Case H, $\sim 5\sigma_{q_v}$ for Cases L-L and H-L, and $\sim 6\sigma_{q_v}$ for Cases L-H. Anderson et al.,⁴⁵ in their experimental investigation of cloud processes, quantified that the maximum T' is $\sim 4\sigma_T$ and maximum q'_v is $\sim 6\sigma_{q_v}$ in the updraft region of cloud. Agreement of the present simulations with the experimental results in Ref. 45 is therefore evident.

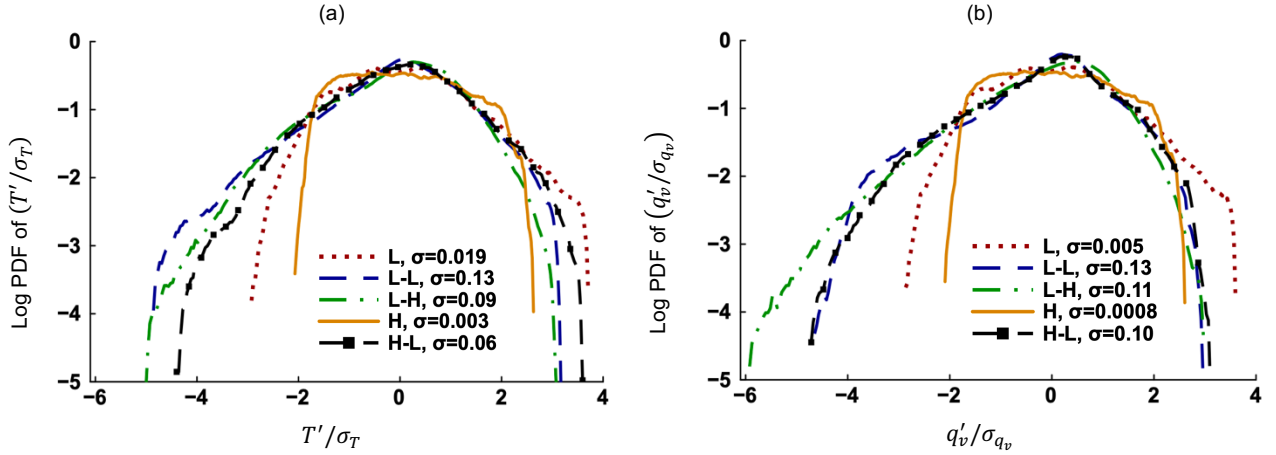


FIG. 8 Probability density functions (PDF) of the (a) fluctuating temperature normalized by the standard deviation (T'/σ_T) and (b) fluctuating water vapor mixing ratio normalized by the standard deviation (q'_v/σ_{q_v}). The ordinate is on a base-10 logarithmic scale. The results correspond to Scenario II at 25 s. The PDFs of (S'_e/σ_{S_e}) resemble that of (q'_v/σ_{q_v}) and hence are not shown. The legends show the values of respective standard deviations.

B. The impact of thermodynamic forcing on microphysics

1. Scenario I

The temporal changes of the mean radii ($\langle r \rangle$), standard deviation (σ_r), and relative dispersion ($\sigma_r/\langle r \rangle$) of all particles (aerosol and cloud) are shown in Figs. 9(a)-(c), respectively. The $\langle r \rangle$ increases exponentially at early times for all cases (Fig. 9(a)), corresponding to positive supersaturation in Fig. 2(c)). However, particle growth levels off at

approximately 25 s for the thermodynamically unforced cases (L and H) while the growth becomes negative for the forced cases (L-L, L-H, and H-L) after 22 s. Thus, condensational growth is followed by evaporative decay if thermodynamic forcing is applied and the decay is faster for a high level of thermodynamic forcing (L-H). The σ_r behaves differently and increases at early times for all five cases. Then it decreases with time for thermodynamically unforced cases (L and H) until becoming statistically stationary (Fig. 9(b)). In the presence of thermodynamic forcing, the σ_r decreases for some time and increases again, with details pending on the differently forced cases. The relative dispersion of particle radii ($\sigma_r/\langle r \rangle$) reaches a peak before gradually leveling off and attaining statistical stationarity if thermodynamic forcing is absent (Fig. 9(c)). For thermodynamically forced cases (L-L, L-H, and H-L), the $\sigma_r/\langle r \rangle$ increases first. Then it decreases before increasing again with time. The $\sigma_r/\langle r \rangle$ is larger and thus the distribution of particle radii is broader with thermodynamic forcing. The behavior of relative dispersion is similar to that of standard deviation, with larger differences at later times due to the simultaneous increase of σ_r and decrease of $\langle r \rangle$. Numerical models often cannot produce particle size distributions as broad as that observed in experiments.⁴⁶ This is because turbulence-microphysics interactions are poorly represented in models, causing the aforesaid discrepancy (Liu et al.⁴⁷). Our results reinforce the previous studies and clarify the role of thermodynamic forcing on the broadening of particle size distributions.

Furthermore, thermodynamic forcing influences the activation and deactivation processes. As shown in Fig. 9(d), the total number of aerosol particles drops to near zero almost instantaneously for all cases after 2 s because all aerosol particles grow beyond the critical radius of $1.26 \mu m$ and activate into cloud droplets. However, careful inspection (zoom-in insert) reveals that some of the previously activated cloud droplets deactivate back into aerosols for two forced cases (L-L and L-H) at late times, unlike the cases without thermodynamic forcing. This corresponds to the evaporative decay reported in Fig. 9(a). These results suggest that although most cloud droplets remain activated, thermodynamic forcing could weaken activation.

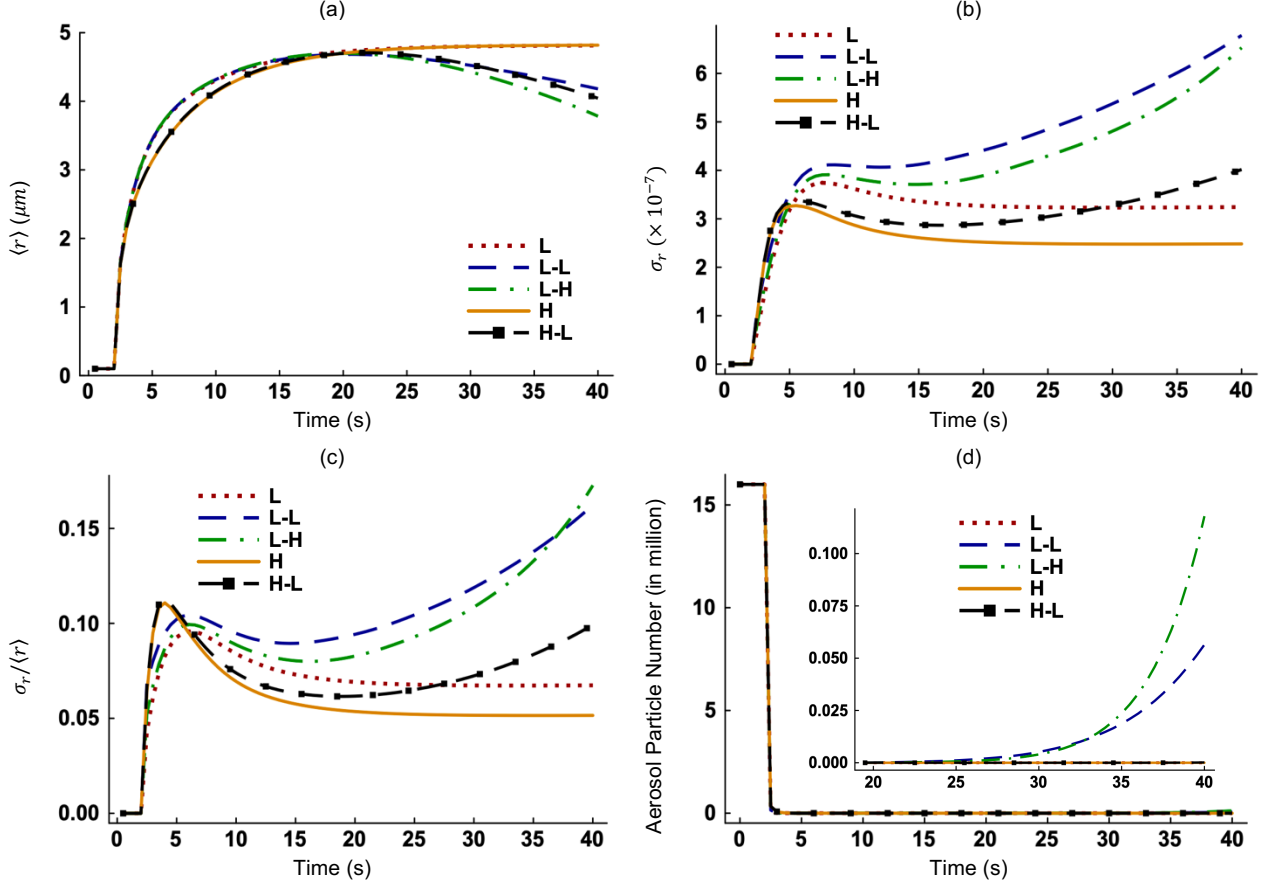


FIG. 9 Time evolution of the (a) mean radius ($\langle r \rangle$) where r includes both aerosol particles and cloud droplets, (b) standard deviation of particle radii (σ_r), (c) relative dispersion of particle radii ($\sigma_r / \langle r \rangle$), and (d) total number of aerosol particles in Scenario I. These results are produced using the Lagrangian data. The ordinate values do not change until particles are allowed to grow after 2 s.

To understand the complex influences of thermodynamic forcing on microphysics, we compare the particle size distributions (PSD) under different forcing conditions and at selected times in Figs. 10(a)-(d). Also shown for completeness is the PSD of monodisperse dry aerosol radii of $0.1 \mu\text{m}$ (dotted green vertical line) that exist initially and the PSD of the corresponding critical radii of $1.26 \mu\text{m}$ (dashed purple vertical line). We see that all the aerosol particles have activated into cloud droplets by 10 s for Case L (Fig. 10(a)) and the PSD appears to be broadening and negatively skewed. By 25 s, the PSD shifts to the right as cloud droplets grow bigger and the distribution narrows. No significant growth takes place between 25 s and 35 s, and the PSDs are identical (Fig. 10(a)). Chen et al.⁴⁸ has referred to this behavior as convergence of PSDs. These results are consistent with the evolution of microphysical properties shown in Fig. 9. With increase of flow turbulence (Case H, Fig. 10(d)), the PSD is narrower at 35 s compared to that of Case L, even though the mean radii are virtually identical for both cases at that time (Fig. 9(a)). The narrowing of PSD suggests that mixing becomes more homogeneous as turbulence intensity increases. In contrast, we see in Figs.

10(b)-(c) that thermodynamic forcing (Cases L-H and L-L) has a strong broadening effect on the PSD, especially at later times. A comparison of Case L (Fig. 10(a)) and Cases L-H and L-L (Figs. 10(b)-(c)) highlights three differences between the thermodynamically unforced and forced cases. First, although the maximum droplet radius increases marginally, the minimum droplet radius decreases considerably at 25 s and 35 s in the presence of thermodynamic forcing. Second, all particles are larger than the critical radius (r_c) at 10 s but some of them are smaller than r_c at later times for the forced cases (L-L and L-H), which indicates that activation is followed by deactivation. Finally, unlike the thermodynamically unforced cases for which the size distributions become narrower and converge with time, the PSDs for the forced cases broaden with time after a brief narrowing without trend of convergence.

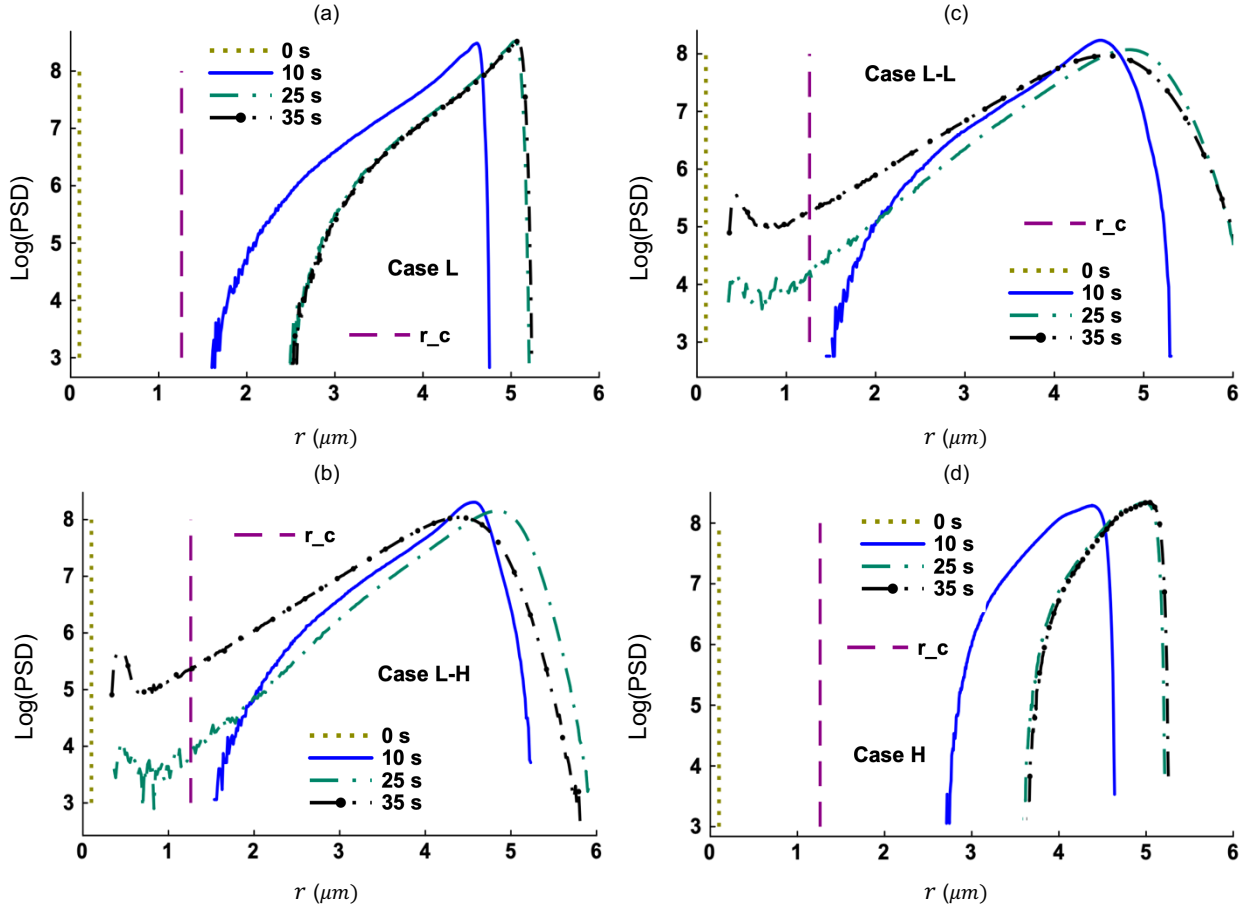
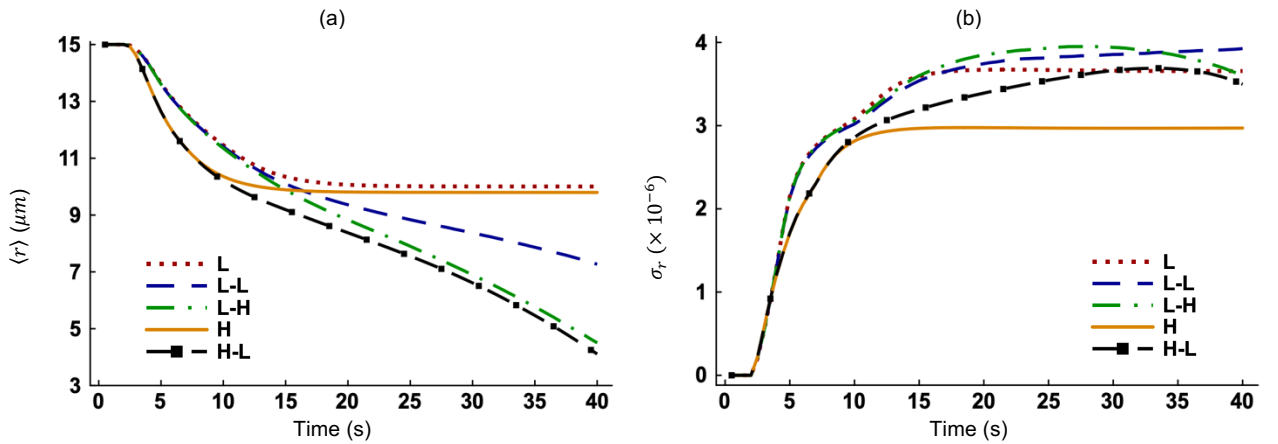


FIG. 10 Particle size distributions (PSD) for (a) Case L, (b) Case L-H, (c) Case L-L, and (d) Case H at selected times in Scenario I. The ordinate is on a base-10 logarithmic scale. The critical radius (r_c) is marked by the dashed vertical line. The r in the abscissa includes both aerosol and cloud particles. The unit of PSD is $\mu\text{m}^{-1}\text{cm}^{-3}$. Note that $\text{PSD} = \text{PDF} \times n_c$, where n_c is the particle number concentration.

2. Scenario II

The modeled cloud droplets have a monodisperse size distribution initially in Scenario II with $15 \mu\text{m}$ radii and experience evaporative decay with time. Figure 11(a) shows that $\langle r \rangle$ decreases faster for three thermodynamically

forced cases (L-L, L-H, and H-L) compared to the unforced cases (L and H). Hence, thermodynamic forcing enhances evaporation in our model. Between Cases L-L and L-H, a high level of thermodynamic forcing (Case L-H) leads to more evaporation (Fig. 11(a)). Evaporation is also enhanced by stronger turbulence and mixing (Cases H vs L). Evaporation increases most when thermodynamic forcing is added to a high level of flow turbulence (Case H-L). The standard deviation increases with time and gradually levels off for the unforced cases (Fig. 11(b)). The changes of the $\langle r \rangle$ and σ_r lead to the unique changes of the $\sigma_r/\langle r \rangle$ in Fig. 11(c). The $\sigma_r/\langle r \rangle$ keeps increasing for the forced cases but becomes statistically stationary otherwise (Cases L and H). These results are qualitatively consistent with Chen et al.⁴⁸, who used a particle-based DNS model without thermodynamic forcing and obtained statistical stationarity for the $\sigma_r/\langle r \rangle$ during turbulent entrainment and mixing. Thermodynamic forcing promotes partial evaporation, for which cloud droplets exist in various sizes. Whereas the domain becomes saturated ($\langle S_e \rangle \sim 0$) in the absence of forcing, making particle growth and its variation (σ_r) stagnant. The number of cloud droplets diminishes over time (Fig. 11(d)) as evaporation and deactivation continues. The reduction is larger for Case L compared to Case H, implying that a higher level of flow turbulence suppresses deactivation in our model. With thermodynamic forcing present, higher turbulence again corresponds to fewer deactivated cloud droplets (Cases L-L vs H-L). The order of deactivation (L-H > H-L and L > H) in Fig. 11(d) is apparently opposite to the order of evaporation (L-H < H-L and L < H) in Fig. 11(a). This is because localized evaporation can cause more droplets to fall below the critical barrier and deactivate, even if the droplet population evaporates less. The converse is also possible. The advantage of the Lagrangian method is evident since it distinguishes between the growth histories of individual droplets.



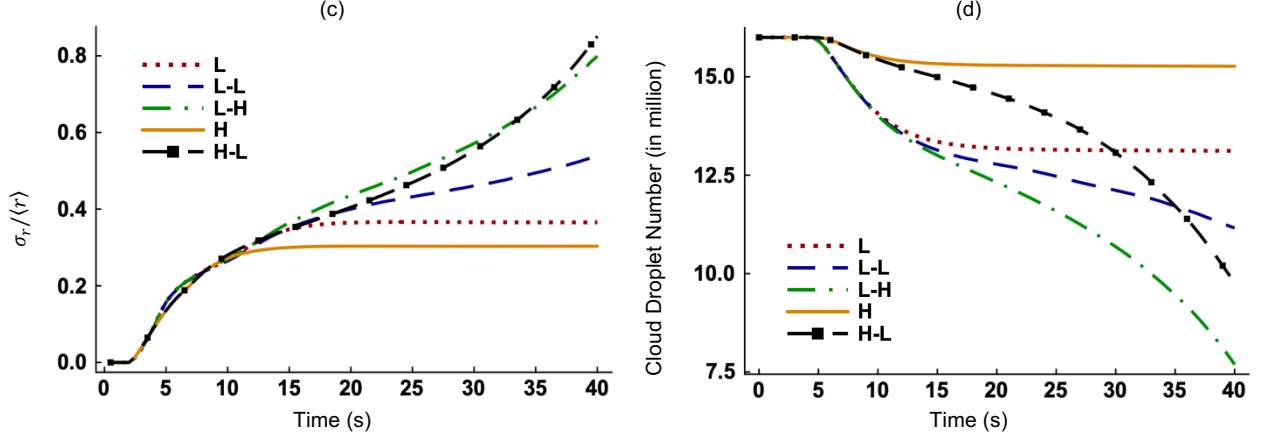
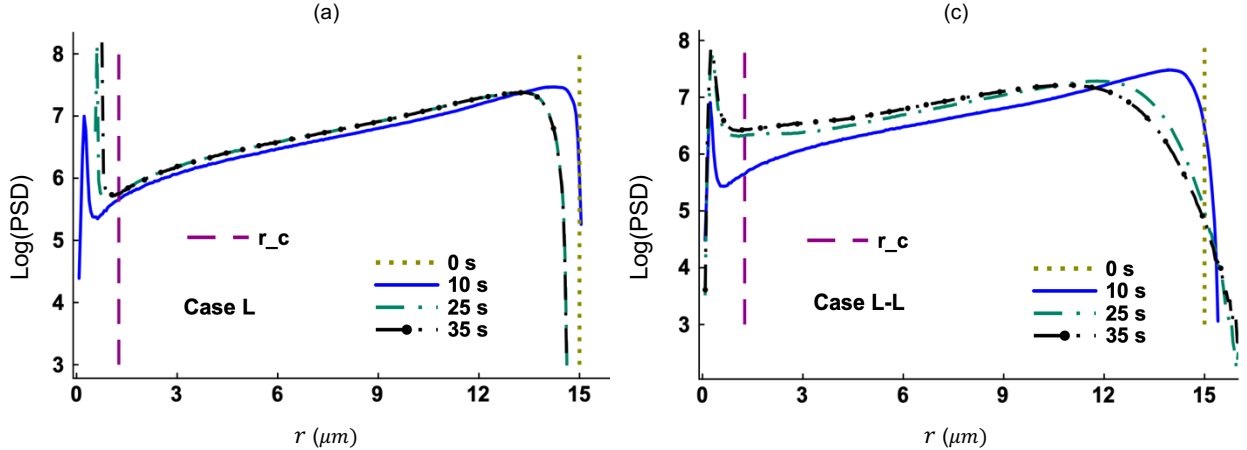


FIG. 11 Time evolution of the (a) mean radius ($\langle r \rangle$) of all particles (aerosol + cloud), (b) standard deviation of particle radii (σ_r), (c) relative dispersion of particle radii ($\sigma_r / \langle r \rangle$), and (d) total number of cloud droplets in Scenario II. These results are produced using the Lagrangian data. The ordinate values do not change until particles are allowed to grow after 2 s.

The PSDs in Scenario II appear to be bimodal for all cases (Figs. 12(a)-(d)), with one peak to the left of the r_c line and another to the right. The existence of bimodality⁴⁹ indicates inhomogeneous evaporation. A group of cloud droplets experiences strong evaporation, becomes much smaller, and deactivates into aerosol particles. These particles correspond to the left peak while the right peak is formed by the rest of the droplet population. Manton⁴¹ suggests that flow turbulence causes cloud droplets to grow (or shrink) differently, creating a bimodal size distribution as shown by our model. For Case L, some droplets evaporate very little by 35 s and their sizes are very close to the initial size of $15 \mu m$ (Fig. 12(a)). If thermodynamic forcing is applied (Case L-L), a small number of cloud droplets grows instead of shrinking (Fig. 12(c)). This makes the PSD broader compared to the unforced case (L). With increased turbulence intensity (Case H), the PSD is narrower (Fig. 12(b)). If thermodynamic forcing is added to a high turbulence condition (Case H-L), more cloud droplets undergo complete evaporation and the maximum droplet radii decrease at all three times shown in Fig. 12(d).



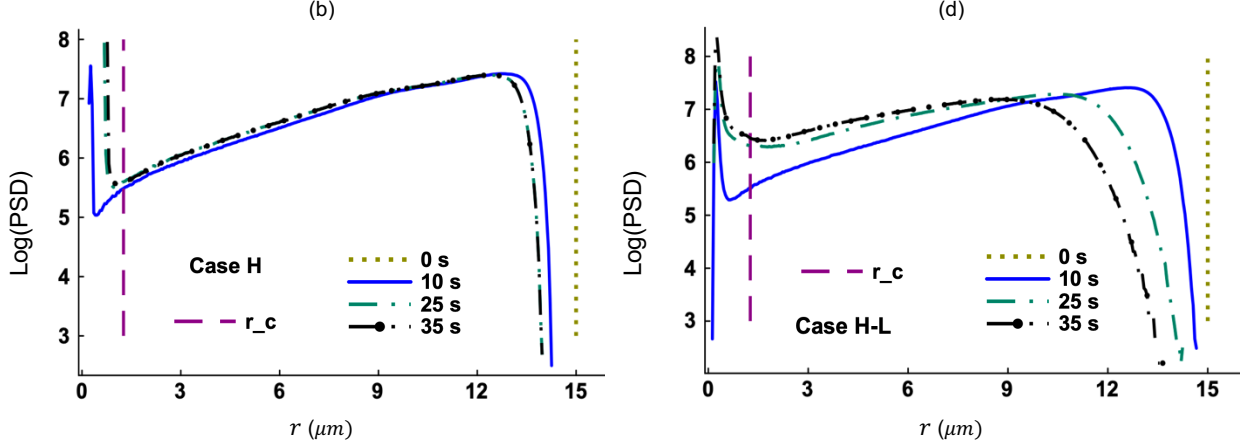


FIG. 12 Particle size distributions (PSD) for (a) Case L, (b) Case H, (c) Case L-L, and (d) Case H-L at selected times in Scenario II. The ordinate is on a base-10 logarithmic scale. The critical radius (r_c) is marked by the dashed vertical line. The r in the abscissa includes both aerosol and cloud particles. The unit of PSD is $\mu\text{m}^{-1}\text{cm}^{-3}$.

C. Further discussion on activation and deactivation

Prabhakaran et al.⁵⁰ identified three regimes of activation (and deactivation) which are i) mean-dominated, ii) fluctuation-mediated, and iii) fluctuation-dominated. In our previous study⁶ that excluded thermodynamic forcing, only mean-dominated regimes were identified. However, as alluded earlier, consideration of thermodynamic forcing leads to activation and deactivation in all three regimes in both Scenarios I and II. Thus, in-depth analysis is conducted here on this understudied topic.

We observe these regimes at different stages of simulation for thermodynamically forced cases (L-L, L-H, and H-L) in Scenario I and discuss the results for Case L-H as example. Figures 13(a)-(c) show the PDFs of environmental supersaturation S_e . The critical supersaturation S_c and the $\langle S_e \rangle$ are also pointed out. At 5 s, mean-dominated activation (MDA) takes place because $\langle S_e \rangle$ is much larger than S_c . At 15 s, fluctuation-dominated deactivation (FDD) and fluctuation-mediated activation (FMA) take place. Although $\langle S_e \rangle$ is slightly larger than S_c , fluctuations make some S_e values smaller than S_c and lead to FDD. FMA occurs when S_e is larger than S_c due to fluctuations. We see that $\langle S_e \rangle$ is slightly smaller than S_c at 25 s and activation is possible only through fluctuations. This translates into fluctuation-dominated activation (FDA). Fluctuation-mediated deactivation (FMD) occurs when $\langle S_e \rangle$ is smaller than S_c and fluctuations make S_e even smaller.

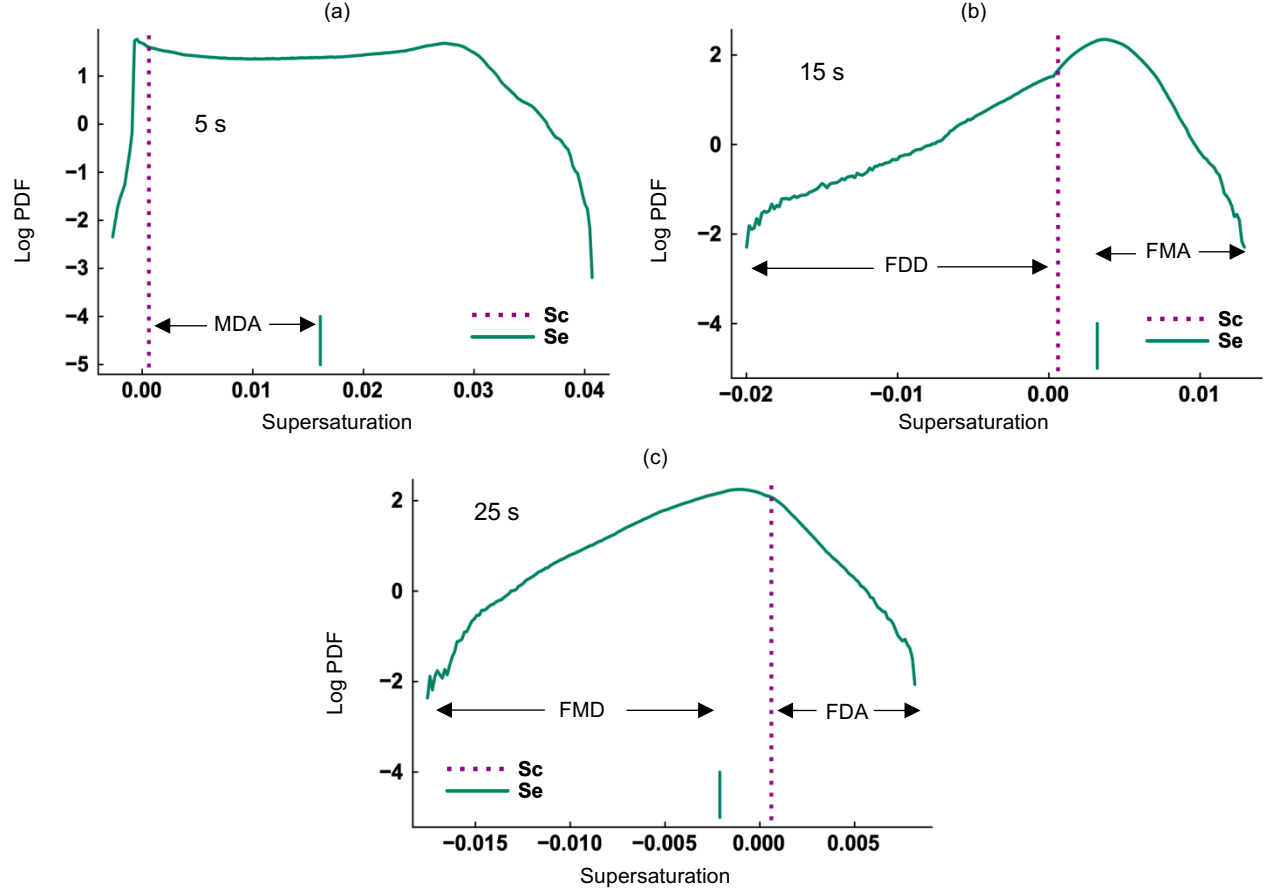
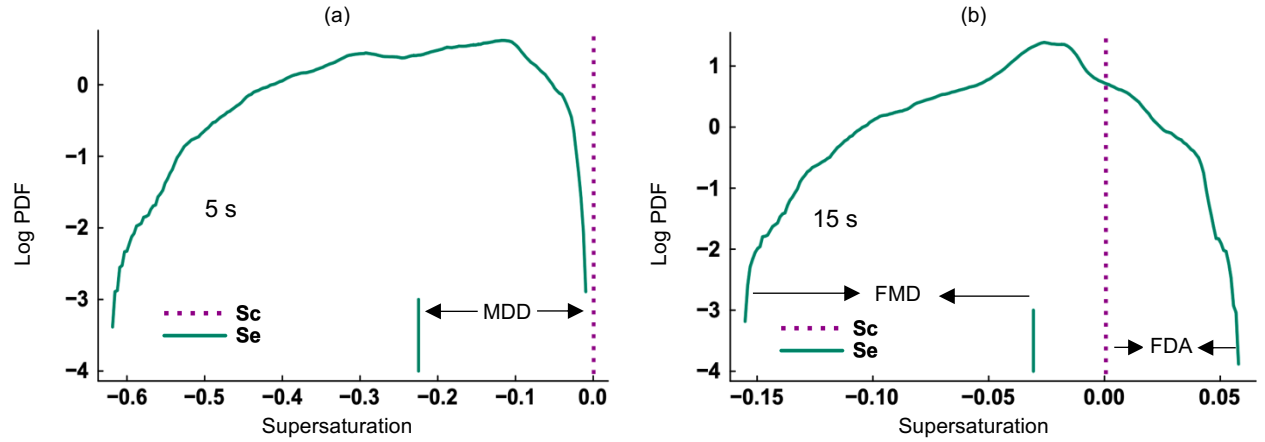


FIG. 13 Log PDFs of the environmental supersaturation (S_e) for Case L-H in Scenario I at (a) 5 s, (b) 15 s, and (c) 25 s. The dotted vertical line shows the value of critical equilibrium supersaturation (S_c). The solid vertical line indicates the mean (expected) S_e . MDA refers to mean-dominated activation, FDD is fluctuation-dominated deactivation, FMA is fluctuation-mediated activation, FMD is fluctuation-mediated deactivation, and FDA imply fluctuation-dominated activation.



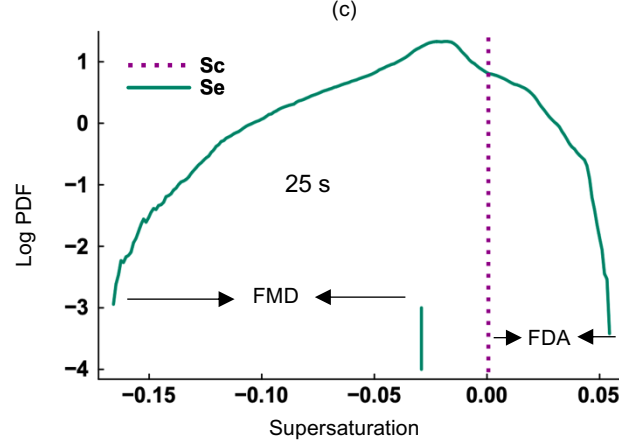


FIG. 14 Log PDFs of the environmental supersaturation (S_e) for Case H-L in Scenario II at (a) 5 s, (b) 15 s, and (c) 25 s. The dotted vertical line shows the value of critical equilibrium supersaturation (S_c). The solid vertical line indicates the mean (expected) S_e . MDD refers to mean-dominated deactivation, FMD is fluctuation-mediated deactivation, and FDA imply fluctuation-dominated activation.

Figures 14(a)-(c) show the PDFs of S_e for Case H-L in Scenario II. The S_c and $\langle S_e \rangle$ are indicated as well. At 5 s, mean-dominated deactivation (MDD) occurs because $\langle S_e \rangle$ is much smaller than S_c . At 15 s and 25 s, $\langle S_e \rangle$ remains smaller than S_c . But fluctuations make some S_e values larger than S_c and support fluctuation-dominated activation (FDA). Fluctuation-mediated deactivation (FMD) takes place when S_e is smaller than S_c because of fluctuations. The results emphasize that fluctuations in the S_e play an important role in determining the inter-conversion of aerosol particles and cloud droplets.

V. CONCLUDING REMARKS

A three-dimensional particle-based DNS model is employed with inclusion of low-wavenumber forcing in the momentum and thermodynamic fields.⁵¹ The influences of thermodynamic forcing on various thermodynamic and microphysical properties are thoroughly examined. The key findings from our simulations are itemized below.

- Thermodynamic forcing reduces the mean temperature, mean vapor mixing ratio, and mean supersaturation during condensation (Figs. 2(a)-(c)) and evaporation (Figs. 2(d)-(f)). Strong flow turbulence or higher level of momentum forcing erases mean supersaturation gradient faster.
- As expected, the fluctuating thermodynamic fields become statistically stationary (Figs. 3(a)-(d)) if external thermodynamic forcing is applied but decay otherwise. The decay is caused by the lack of energy to maintain thermodynamic fluctuations. A higher level of momentum forcing accelerates the decay through stronger mixing and homogenization.

- Although thermodynamic forcing is applied at large scales only, it affects the energy spectra of thermodynamic fields at all scales (Figs. 4(a)-(b)) by interacting with the phase change term. The back transfer of energy from small to large scales is evident in the spectra of particle radius (Fig. 5(c)) and effective diffusivity (Fig. 6(b)).
- Velocity fluctuations do not have significant impact on supersaturation fluctuations in our model (Fig. 7(c)). External thermodynamic forcing is the primary source of water vapor fluctuations (Fig. 7(b)).
- The PDFs of fluctuating temperature and water vapor fields are negatively skewed for the thermodynamically forced cases but positively skewed for the unforced cases (Figs. 8(a)-(b)). The kurtosis of temperature and vapor PDFs are greater than 3 for the forced cases but less than 3 for the unforced cases.
- Condensation and activation in Scenario I are followed by evaporation and deactivation if thermodynamic forcing is present (Figs. 9(a) and 9(d)). The relative dispersion of particle radii ($\sigma_r/\langle r \rangle$) becomes statistically stationary without forcing but keeps increasing with forcing (Fig. 9(c)).
- The particle size distributions become narrower during condensation if the turbulence intensity increases and no thermodynamic forcing is applied (Fig. 10(d)). With thermodynamic forcing, a small fraction of the previously activated cloud droplets deactivates back into aerosol particles, and the PSDs broaden (Figs. 10(b)-(c)).
- Thermodynamic forcing enhances both evaporation (Fig. 11(a)) and deactivation (Fig. 11(d)). A high level of momentum forcing enhances domain-averaged evaporation (Fig. 11(a)) but suppresses deactivation of individual cloud droplets (Fig. 11(d)).
- The particle size distributions are found to be bimodal during evaporation (Figs. 12(a)-(d)). They are broader in response to thermodynamic forcing if the turbulence intensity is low (Fig. 12(c)). More cloud droplets evaporate completely if the turbulence level is high and thermodynamic forcing is present (Fig. 12(d)).
- Mean-dominated activation occurs at 5 s in Scenario I (Fig. 13(a)) and it transitions to fluctuation-dominated deactivation and fluctuation-mediated activation at 15 s (Fig. 13(b)). These regimes are followed by fluctuation-dominated activation at 25 s (Fig. 13(c)). Mean-dominated deactivation takes place at early times in Scenario II (Fig. 14(a)) and is followed by fluctuation-mediated deactivation and fluctuation-dominated activation at later times (Fig. 14(b)).

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DATA AVAILABILITY

The data supporting the findings of this work are available from the corresponding author upon reasonable request.

APPENDIX A: THE IMPACT OF ADIABATIC LAPSE RATE

Previous DNS models such as those of Gao et al.⁵ and Kumar et al.³ ignored the adiabatic lapse rate term $(-g/c_p)w$ in the temperature transport equation because the rate of cooling with increasing altitude is insignificant if the domain size is small. To assess the impact of this adiabatic lapse rate term, we modify Eq. (5) as:

$$\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla)T = \frac{L_h}{c_p} C_d + \mu_T \nabla^2 T - \frac{g}{c_p} w. \quad (A1)$$

Figures 15(a)-(d) show the temporal evolution of the $\sigma_{S_e}^2$, $\langle S_e \rangle$, $\langle r \rangle$, and $\sigma_r / \langle r \rangle$ for Cases L, L-L, and L-LR in Scenario I. Case L-LR solves Eq. (A1) instead of Eq. (5) and does not include water vapor forcing. We see that the decay of supersaturation variance is prevented for Case L-LR (Fig. 15(a)) but $\sigma_{S_e}^2$ is smaller than that for Case L-L by orders of magnitude. The profiles of $\langle S_e \rangle$, $\langle r \rangle$, and $\sigma_r / \langle r \rangle$ are identical for Cases L and L-L (Figs. 15(b)-(d)). Thus, adiabatic lapse rate does not influence cloud microphysics in our DNS model. Therefore, Fourier forcing (Case L-L) is necessary to effectively maintain thermodynamic fluctuations and have desired variation in microphysical properties.

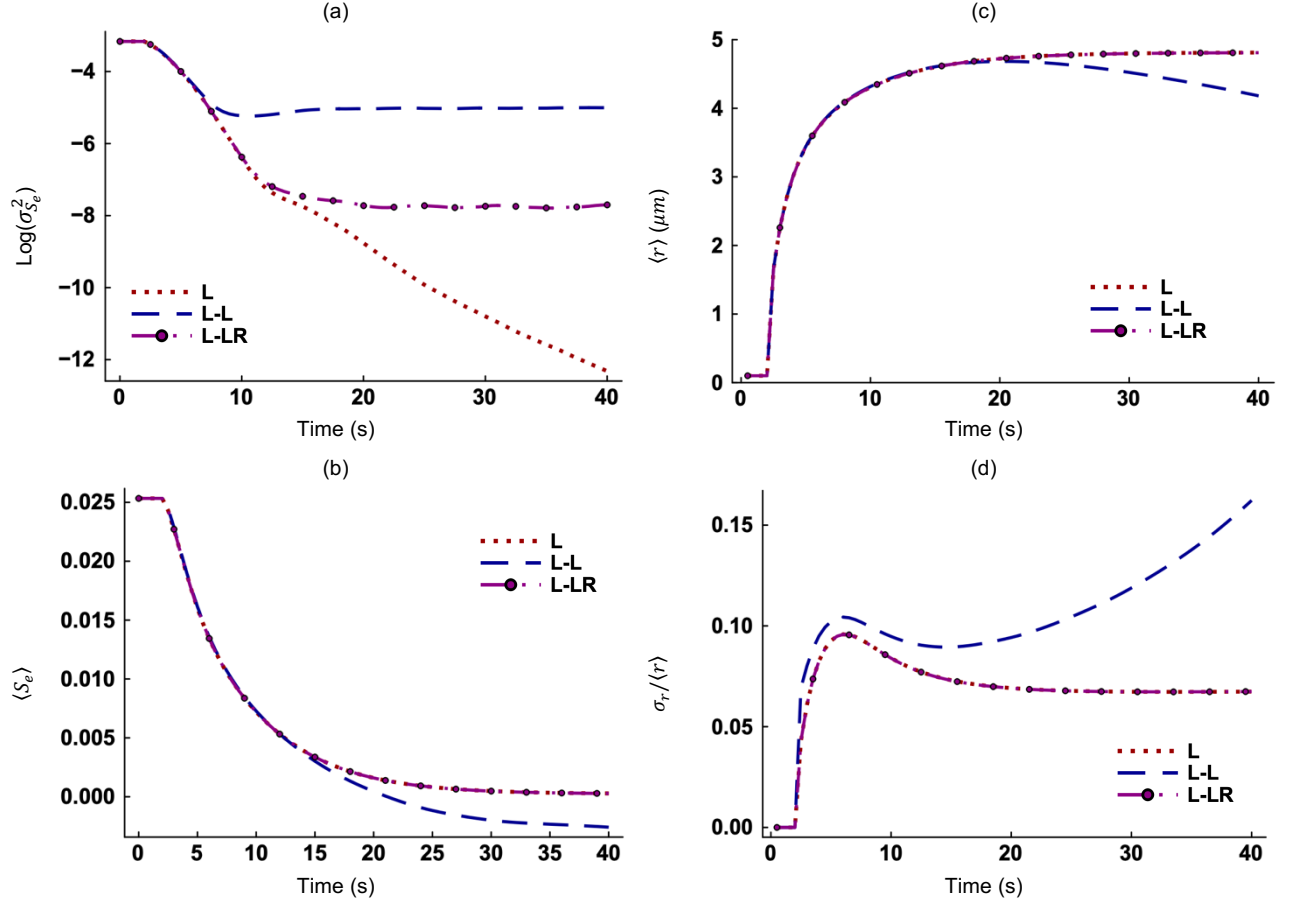


FIG. 15 Time evolution of the (a) variance of environmental supersaturation, (b) mean environmental supersaturation, (c) mean radius, and (d) relative dispersion of particle radii. The results represent Scenario I.

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